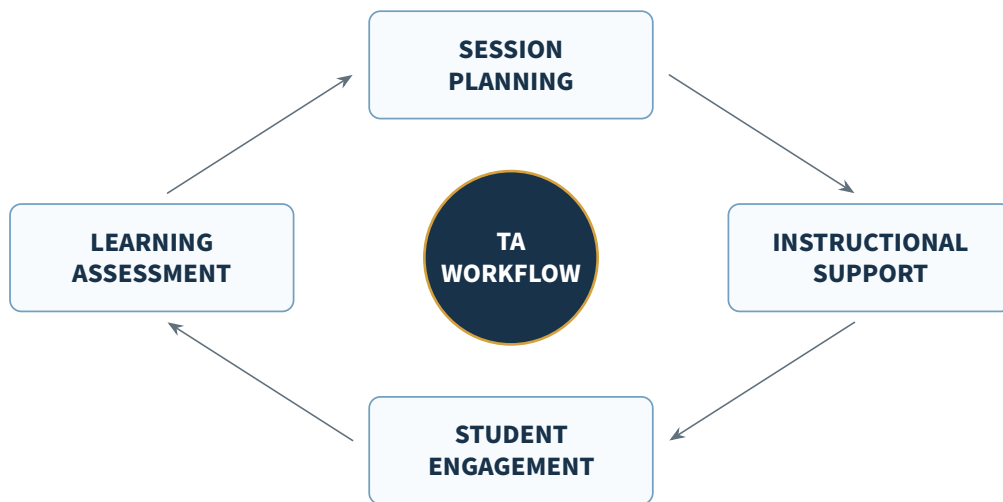




Enterprise Resource Planning

Teaching Assistant Guide



COURSE OUTLINE

ERP concepts, business processes, and integrated Odoo 19 Community Edition practicals for West End Bicycles.

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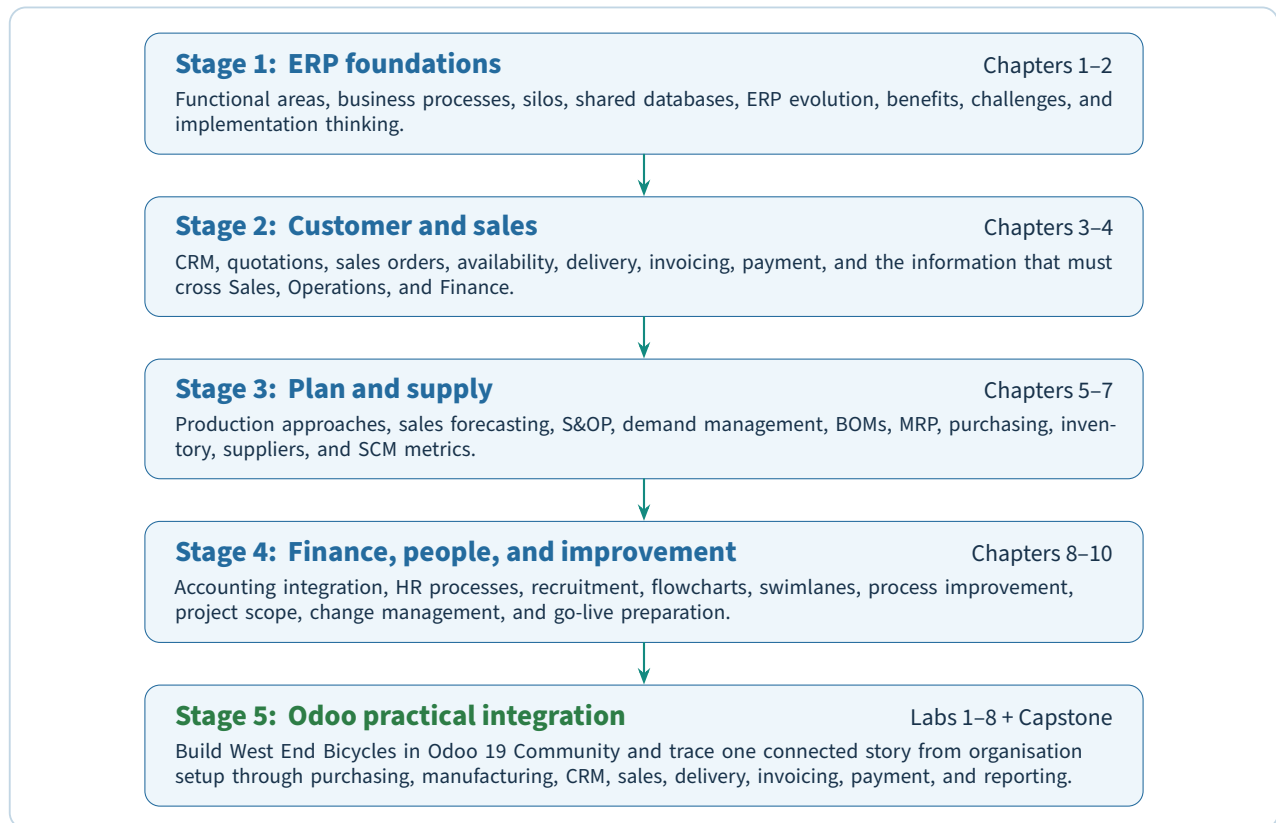


Learning Path

Course Map: Follow the Flow of Information	ii
Stage 1 ERP Foundations and Business Processes	1
Chapter 1 Business Functions, Business Processes, and ERP Integration	2
Chapter 2 The Development of ERP Systems: Architecture, Benefits, and Challenges	6
Stage 2 Customer, Sales, and Revenue	11
Chapter 3 Marketing Information Systems and the Sales Order Process	12
Chapter 4 Customer Relationship Management	17
Stage 3 Planning, Production, and Supply Chain	21
Chapter 5 Production Planning: From Demand to a Feasible Production Plan	22
Chapter 6 Sales Forecasting and Sales & Operations Planning	27
Chapter 7 Supply Chain Management, MRP, Purchasing, and Inventory	31
Stage 4 Finance, People, and Process Improvement	36
Chapter 8 Accounting and Finance in an ERP System	37
Chapter 9 Human Resources Processes and Recruitment	41
Chapter 10 Process Modeling, Process Improvement, and ERP Implementation	46
Stage 5 Integrated Odoo Practicals	51
Lab 1 of 8 — Business Process and Odoo Environment Setup	52
Lab 2 of 8 — ERP Benefits, Challenges, and Modular Integration	55
Lab 3 of 8 — Production Planning and Manufacturing in Odoo	57
Lab 4 of 8 — Customer Relationship Management in Odoo	59
Lab 5 of 8 — Sales Order Process: Order-to-Cash in Odoo	61
Lab 6 of 8 — Sales Forecasting and S&OP with ERP Data	63
Lab 7 of 8 — Supply Chain Management: Procure, Receive, and Measure	65
Lab 8 of 8 — Human Resources and Recruitment Process Modelling	67
Capstone: West End Bicycles End-to-End ERP Story	69
Appendix A: Odoo 19 Community Quick Reference	73
Appendix B: Course Source and Tutorial Map	76

Course Map: Follow the Flow of Information

ERP becomes much easier when you stop treating each module as an isolated topic. The course follows the same logic as a real business: understand the organisation, capture customer demand, plan and obtain resources, fulfil the order, record the financial consequences, and improve the process.



Key idea

The central question throughout the course is not “Which screen do I click?” It is “Which business event happened, what data does that event create, which functional areas need the data, and what should the ERP system do next?”

Stage 1

ERP Foundations and Business Processes

Business Functions, Business Processes, and ERP Integration

Chapter at a glance

Before you start: You only need a basic idea of what departments such as Sales, Finance, Supply Chain, and HR do.

By the end, you can: distinguish a functional area from a business function and a business process, trace cross-functional information flows, and explain why a shared ERP database improves coordination.

Source basis

This chapter follows Tutorial 1 on Business Process / Business Functions and Chapter 1 of *Concepts in Enterprise Resource Planning*. The West End Bicycles scenario and the four functional areas are preserved from the course materials; the explanations and diagrams below reorganise those ideas into a study sequence.

1.1 Start with the organisation, not the software

An ERP system exists to support how a business works. Before learning menus and modules, identify the work performed by the organisation.

A **functional area** is a broad part of the organisation responsible for a family of activities. The course materials emphasise four recurring areas:

Functional area	Typical business functions
Marketing and Sales	Product promotion, pricing, quotations, sales orders, customer support, CRM, and sales forecasting.
Supply Chain Management	Purchasing, receiving, inventory, production planning, manufacturing, logistics, and delivery.
Accounting and Finance	Recording transactions, receivables and payables, budgeting, cash management, financial reporting, and profitability analysis.
Human Resources	Recruiting, hiring, training, payroll-related administration, benefits, employee records, and compliance.

A **business function** is one activity within a functional area. For example, “taking a sales order” is a business function in Marketing and Sales. A **business process** is broader: it is a connected set of activities that transforms an input into an output that matters to a customer or another internal user.

Plain idea

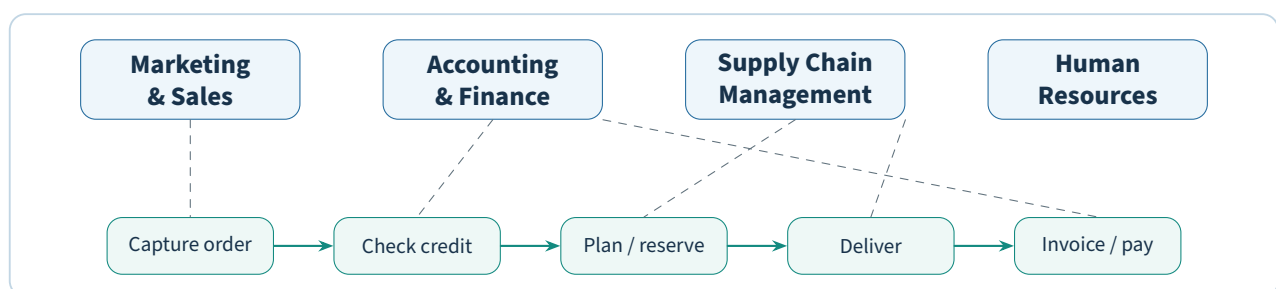
A department is an organisational box. A process is a journey. The customer order journey may begin in Sales, move through credit checking, inventory, delivery, and invoicing, and end when payment is recorded. The customer sees one experience even though the company sees several departments.

1.2 Why business processes cross functional boundaries

Suppose West End Bicycles receives an order for 1,000 bicycles. The sales employee can capture the request, but Sales alone cannot complete the order. The organisation must answer several questions:

- Is the customer allowed to buy on the requested credit terms?
- Are enough bicycles already in stock?
- If not, can production make them on time?
- Are the required frames, tyres, brakes, gears, and chains available?
- When can the order be delivered?
- What invoice should Finance issue, and has the customer paid?

Each question belongs partly to a different functional area. The process therefore crosses functional lines.



The HR area may not touch every individual order, but it still supports the process by ensuring that the organisation has trained salespeople, buyers, production planners, accountants, and warehouse staff.

1.3 The information each area needs from the others

A process can fail even when every department works hard if information moves slowly or inconsistently.

1.3.1 Marketing and Sales

Marketing and Sales needs customer details, order data, product availability, unit cost and profitability information, and staffing support. It produces sales orders, prices, sales forecasts, and customer demand information.

1.3.2 Supply Chain Management

SCM needs sales forecasts and confirmed orders so it can plan production, purchasing, and inventory. It produces availability, production, receipt, and shipment information that Sales and Finance need.

1.3.3 Accounting and Finance

Finance needs sales, purchasing, inventory, production, payroll, and payment data. It produces invoices, credit information, payments to suppliers, financial statements, and management reports.

1.3.4 Human Resources

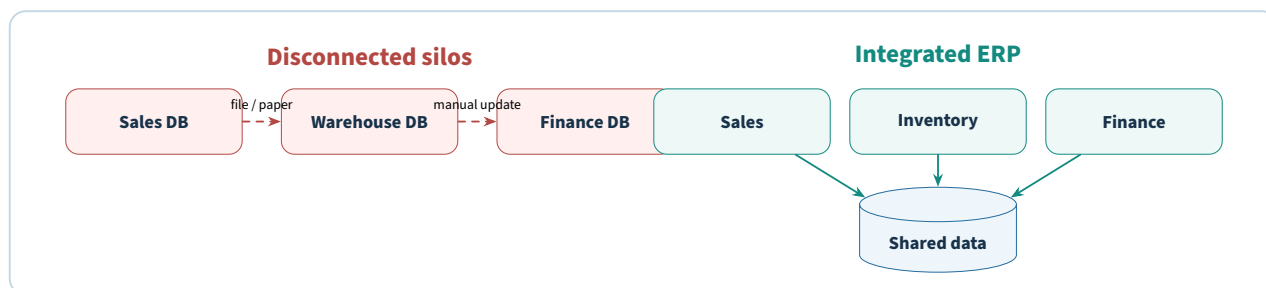
HR needs staffing requirements and job-skill information from operational departments. It produces employee records, training information, compliance records, and compensation-related data.

Key idea

The quality of a decision is constrained by the quality, timeliness, and consistency of the data available when the decision is made. ERP integration is valuable because it reduces the delay and disagreement created by multiple disconnected copies of the same business facts.

1.4 Silos versus an integrated information system

A **silo** is a departmental information system that stores its own data and does not automatically stay synchronised with other systems. A silo is not automatically bad; the problem appears when a cross-functional process depends on information that exists in several different versions.



With a shared database, a business event can update several views of the company at once. For example, when a delivery is validated, Inventory sees stock leave the warehouse, Sales sees fulfilment progress, and Finance has the evidence needed to bill or recognise the related transaction according to the configured process.

1.5 West End Bicycles as the running example

The tutorial scenario describes West End Bicycles as a manufacturer of road, mountain, and city bicycles, along with accessories such as helmets, shoes, and clothing. That scenario is useful because it requires all four functional areas.

Worked example

For a Road Bike order, the process can be read as two connected cycles:

1. **Procure-to-pay:** identify the need for frames, tyres, and parts; create purchase orders; receive materials; update inventory; pay suppliers.
2. **Order-to-cash:** capture customer interest; create a quotation; confirm a sales order; deliver bikes; create an invoice; receive payment.

Manufacturing connects the two cycles by converting purchased components into finished Road Bikes.

1.6 What Odoo adds to this picture

The supplied Odoo 19 Community guide implements the same concepts using eight applications: Contacts, CRM, Employees, Inventory, Invoicing, Manufacturing, Purchase, and Sales.

Odoo 19 Community

The Odoo guide emphasises three ideas that will recur throughout the practicals:

- Records are linked. A purchase order leads to a receipt; a sales order leads to a delivery and an invoice.

- Business documents move through states such as draft, confirmed/posted, and done/paid.
- Smart buttons expose the relationships between records, making the information flow visible instead of requiring separate manual reconciliation.

Common mistake

Do not define ERP as merely “software that combines departments.” The important point is process integration: the same underlying business event should be visible consistently to the functions that need it, at the time they need it.

Check your understanding

1. Explain the difference between a functional area, a business function, and a business process using West End Bicycles.
2. Why can a weekly printed credit report cause a sales-order problem even if the Accounting data was correct when the report was printed?
3. Name three pieces of information that must move between Sales and Supply Chain when fulfilling a bicycle order.

Chapter recap

- Functional areas organise responsibility; business processes connect activities across those areas.
- Marketing and Sales, SCM, Accounting and Finance, and HR depend on one another’s data.
- Disconnected systems create duplicate entry, delays, inconsistent data, and poor visibility.
- ERP uses integrated data and linked processes so that one business event can support several departments without re-entering the same facts.
- Odoo will be used in the practicals to make those information flows visible through linked records and document states.

The Development of ERP Systems: Architecture, Benefits, and Challenges

Chapter at a glance

Before you start: You should understand the idea of functional silos from Chapter 1.

By the end, you can: explain how MRP evolved into ERP, describe ERP modularity and shared data, evaluate major benefits and implementation challenges, and identify conditions that make an ERP project more likely to succeed.

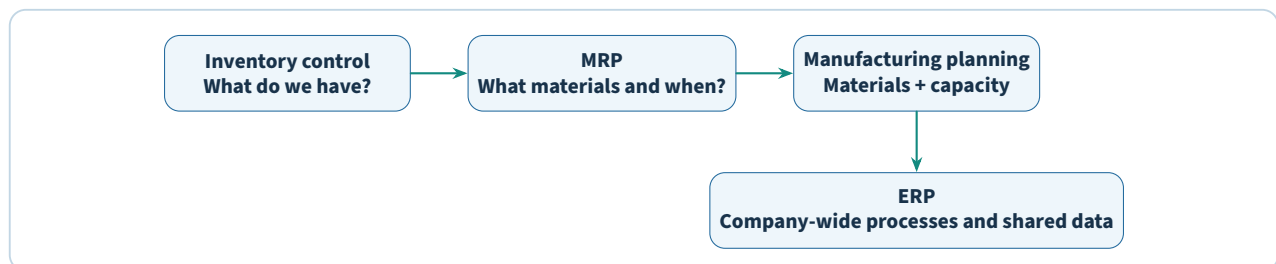
Source basis

This chapter is based on Tutorial 2 and Chapter 2 of *Concepts in Enterprise Resource Planning*, with implementation ideas reinforced by Chapter 7. Tutorial 2 asks students to investigate one ERP benefit and one ERP challenge, support each with evidence, and teach the result to peers.

2.1 Why ERP systems emerged

Business computing did not begin with an integrated company-wide platform. Early information systems were typically built to solve one department's immediate problem. A manufacturing system tracked materials, an accounting system recorded financial transactions, and a sales system captured customer orders. Each system could be useful by itself while still making the company as a whole difficult to coordinate.

The manufacturing lineage is especially important. **Material Requirements Planning (MRP)** focused on calculating what materials were needed and when. Later systems expanded toward manufacturing resource planning and then broader enterprise integration. ERP extended the idea beyond factory planning to include sales, purchasing, finance, human resources, logistics, and other processes in one coordinated environment.



2.2 The core architectural idea: modules around shared data

ERP products are modular because businesses perform different kinds of work. A sales module needs quotations and orders; a manufacturing module needs bills of materials and production orders; finance needs invoices, payments, and accounting records. Modularity lets specialised workflows exist while still sharing the same master and transaction data.

Two types of data are useful to distinguish:

Master data Relatively stable facts reused across many transactions: customers, vendors, products, employees, units of measure, prices, and chart-of-account structures.

Transaction data Events that occur over time: a sales order, purchase order, receipt, manufacturing order, delivery, invoice, or payment.

Key idea

ERP integration is not achieved by putting every task on one giant screen. It is achieved by allowing specialised modules to use common master data and to create linked transaction records from the same business events.

2.3 Real-time information and one version of the business event

In the textbook, **real time** means that data and processes are current rather than waiting for periodic transfers. A confirmed order can therefore be available to the functions that need it without a clerk exporting a file, printing a report, or re-entering the data.

This does not mean every organisation literally updates every metric with zero delay. The more useful operational meaning is that the system is designed so that authorised users work from the same current transaction records rather than from separately maintained departmental copies.

2.4 Major benefits of ERP

The course source highlights several benefits. They can be grouped into four mechanisms.

Mechanism	What changes	Business value
Data integration	People and modules use common records rather than separate departmental copies.	Fewer inconsistencies, duplicate entries, reconciliation tasks, and delays.
Process integration	One transaction can trigger the next step in another function.	Faster order fulfilment, purchasing, billing, and reporting.
Management visibility	Managers can view consolidated operational information without manually assembling reports.	More time can be spent managing and improving operations rather than merely discovering what happened.
Global standardisation	Common processes and data structures can span units, locations, currencies, and languages.	Easier coordination and comparable reporting across the organisation.

Other benefits are often consequences of these mechanisms: better customer service, lower inventory, shorter cycle times, improved reporting, stronger control, and a platform for analytics.

Worked example

Suppose West End Bicycles stores the Road Bike price separately in a salesperson's spreadsheet and in Finance's invoice system. A price change must be updated twice and may be missed in one place. With an integrated ERP record, the quotation and later invoice can use the same transaction data, reducing the chance of contradictory prices.

2.5 Why ERP projects are difficult

ERP software affects many departments at once, so implementation is an organisational change project as much as a technology project. The source material identifies costs such as software, hardware, consultants, project-team time, training, implementation disruption, and productivity losses during transition.

More importantly, failed or difficult implementations are frequently associated with process and people problems:

- assuming new software will repair a fundamentally bad business process without redesigning it;
- insufficient planning and analysis before configuration begins;
- inadequate education and training;
- treating the implementation as an IT-only project instead of giving process owners responsibility;
- weak executive sponsorship;
- poor organisational change management;
- excessive scope or uncontrolled customisation;
- difficult migration of master and open transaction data from legacy systems.

Common mistake

“Install ERP” is not a business objective. A project needs a process objective such as reducing order cycle time, improving inventory accuracy, shortening financial close, or eliminating repeated manual entry. Otherwise the organisation can spend heavily and merely reproduce its old problems inside a newer system.

2.6 Standard processes versus customisation

ERP packages usually embody a particular way of doing business. The implementation team must decide whether to:

1. adapt the company's process to the standard ERP workflow;
2. configure the ERP system using supported options; or
3. customise the software when a real competitive or regulatory need cannot be met by standard configuration.

Customisation is not automatically wrong, but every custom feature increases implementation effort and can make future upgrades harder. A useful question is: *Does this process create competitive advantage, or is it merely familiar?*

2.7 How to judge ERP success

A successful ERP implementation is not simply one that goes live. It should improve measurable business outcomes.

Possible indicators include:

- order-to-delivery cycle time;
- inventory accuracy and stockout frequency;
- invoice and payment processing time;

- days required to close financial accounts;
- forecast accuracy;
- percentage of transactions completed without manual re-entry;
- user adoption and process compliance;
- customer service levels;
- realised cost savings or revenue improvements.

Some ERP benefits are hard to express as immediate cash savings. Better visibility, improved control, consistent data, and the ability to support future growth may still be strategically important.

2.8 Implementation success factors

The source repeatedly returns to a few conditions that increase the chance of success:

Success factor	Why it matters
Executive sponsorship	Gives the project authority, resources, and cross-functional priority.
Clear scope and objectives	Prevents uncontrolled expansion and makes benefits measurable.
Process ownership	Ensures decisions are made by people who understand the work, not only the technology.
Training and communication	Helps users understand both the system and the new process.
Change management	Addresses resistance, role changes, and new responsibilities.
Data preparation	Poor master data can undermine even correctly configured software.
Testing and staged readiness	Finds configuration and process problems before live transactions depend on the system.

2.9 How Odoo illustrates ERP modularity

The West End Bicycles guide uses eight Odoo 19 Community applications: Contacts, CRM, Employees, Inventory, Invoicing, Manufacturing, Purchase, and Sales. Each application has a distinct job, but the records are connected.

Odoo 19 Community

A useful way to read the Odoo app grid is as a modular ERP architecture:

- **Contacts**: shared customer and vendor identities.
- **CRM** and **Sales**: demand, opportunities, quotations, and orders.
- **Purchase**, **Inventory**, and **Manufacturing**: sourcing, stock, and production.
- **Invoicing**: customer invoices and payments in the Community workflow used by the course.
- **Employees**: departments and people behind the processes.

The same product, customer, vendor, and order information can therefore be reused instead of rebuilt independently in each app.

Try it yourself

Choose one ERP benefit and one ERP implementation challenge from this chapter. For each, write: (1) the mechanism that creates the benefit/challenge, (2) one operational example at West End Bicycles, and (3) one metric that could show whether the issue improved after implementation. This mirrors the structure of Tutorial 2 while forcing the argument to be measurable.

Check your understanding

1. What is the difference between master data and transaction data? Give two Odoo examples of each.
2. Why is top-management support important in a cross-functional ERP implementation?
3. Why can copying an inefficient legacy process exactly into an ERP system reduce the expected benefit?

Chapter recap

- ERP grew from narrower planning and departmental information systems into company-wide integrated platforms.
- ERP modules remain specialised but share master data and linked transaction data.
- The main benefits come from integration, visibility, process coordination, and standardisation.
- The main risks involve process design, people, scope, training, data migration, and change management as much as software.
- Success should be measured in business outcomes, not merely by whether the system was installed.

Stage 2

Customer, Sales, and Revenue

Marketing Information Systems and the Sales Order Process

Chapter at a glance

Before you start: You should understand functional integration and the idea of shared transaction data.

By the end, you can: describe an end-to-end sales order process, diagnose problems caused by disconnected systems, explain the order-to-cash sequence, and show how sales, inventory, and finance are linked.

Source basis

This chapter uses Tutorial 5 and Chapter 3 of *Concepts in Enterprise Resource Planning*. The Fitter Snacker case is retained as the main example of an inefficient, unintegrated sales process, while the Odoo sections map the same concepts onto the West End Bicycles practical.

3.1 What the sales order process includes

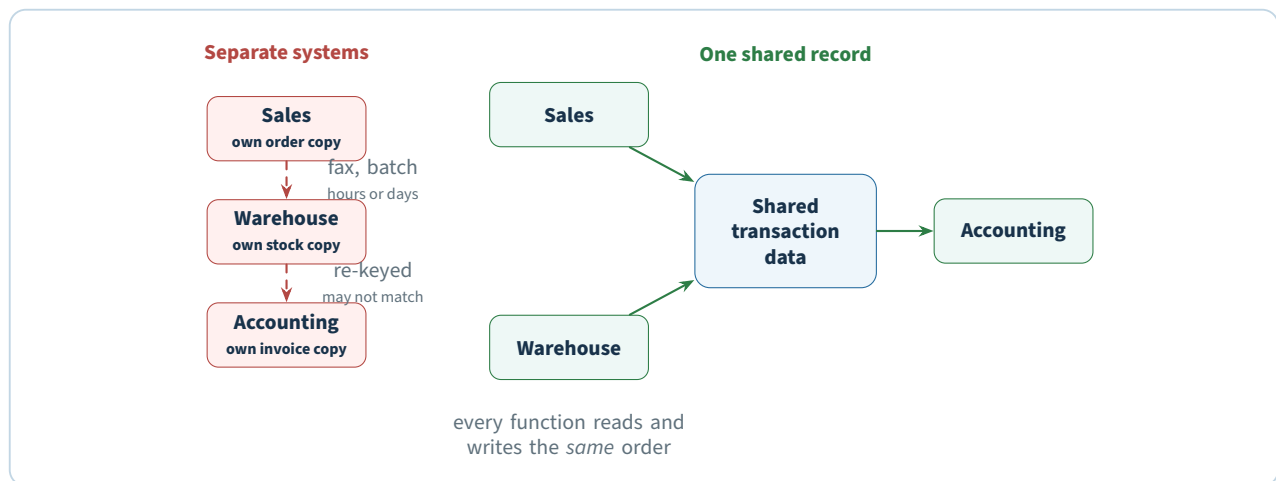
The **sales order process** is the flow of events from a customer's request through fulfilment and payment. It is broader than entering an order. A typical sequence includes quotation, order capture, credit or payment terms, availability checking, picking and delivery, invoicing, payment, and sometimes returns.



Each step creates information needed by another function. If the organisation treats these as isolated departmental tasks, the customer experiences delays and inconsistency.

3.2 Fitter Snacker: what goes wrong without integration

Fitter Snacker runs separate sales, warehouse, and accounting systems. Data crosses between them by fax, paper report, and periodic file transfer. Every one of its recurring problems traces back to a single structural fact: **each department holds its own copy of the order, and the copies drift apart.**



Read the five textbook symptoms as one cause seen from five desks:

Symptom	What actually happens	The stale copy
Quotation and pricing	Arithmetic slips, discounts combined wrongly, terms the office has not seen; faxes misread into duplicate orders.	<i>price</i>
Availability and dates	The clerk asks the warehouse for a ship date built on stock figures that are already out of date.	<i>stock</i>
Credit	The customer is judged against a periodic printed report, so a recent payment is invisible — good customers refused, over-exposed ones approved.	<i>receivables</i>
Order filling	Paper picking lists travel in batches; stock moves that nobody writes down never reach the records.	<i>inventory</i>
Invoicing	Order data is transferred to accounting on a schedule, so fulfilment changes after the transfer are billed wrongly.	<i>shipped quantity</i>

Common mistake

None of this is careless staff; it is process design. A process that requires people to copy, fax, re-key, and reconcile the same fact across systems will produce errors and delays no matter how conscientious they are. Fix the structure, not the people.

3.3 What an integrated ERP changes

An integrated ERP system replaces separate copies of the order with a linked document flow. The same customer, product, price, quantity, delivery, invoice, and payment records can be traced through the process.

A strong sales-order system therefore performs several checks at the time of the transaction:

- validates customer and product master data;
- applies configured pricing and discounts;
- checks inventory or planned availability;
- checks credit or payment conditions where configured;
- creates fulfilment documents from the order;
- passes billing data to Finance;

- records status changes so Sales can answer customer questions.

3.4 Order-to-cash as a cross-functional cycle

The term **order-to-cash** emphasises that the process is not complete when the sales order is confirmed. The business has not earned the operational outcome it wants until the product is fulfilled and the money is collected.

Stage	Primary owner	Information created or consumed
Opportunity / quotation	Sales / Marketing	Customer need, product, quantity, expected price, proposed terms.
Sales order	Sales	Confirmed demand and commercial commitment.
Availability and delivery	Inventory / Operations	Stock reservation, picking, actual shipped quantity, delivery status.
Invoice	Finance	Amount due, tax and payment terms, receivable record.
Payment	Finance	Cash received and settlement status, visible back to Sales.

3.5 Integration of Sales and Accounting

The textbook stresses that in an ERP system accounting consequences can be created from operational events rather than reconstructed later. That matters because management reporting, customer balances, and profitability analysis depend on the same source transaction.

For example, if a customer receives a discount, both Sales and Finance need the same final price. If Finance re-enters the invoice manually, the organisation creates an unnecessary opportunity for disagreement.

Key idea

A reliable sales process preserves document lineage: the quotation explains how the sales order was formed, the delivery explains what was physically fulfilled, the invoice explains what was billed, and the payment explains what was settled.

3.6 How the sequence appears in Odoo 19 Community

The supplied West End Bicycles guide maps the order-to-cash flow to linked Odoo records.

Odoo 19 Community

For the Road Bike example, the practical sequence is:

1. create a CRM opportunity for Seri Iskandar Cycling Club;
2. create a quotation for 5 Road Bikes at RM 1,000 each;
3. confirm the quotation so it becomes a Sales Order;
4. open the linked Delivery and validate shipment of 5 bikes;
5. return to the Sales Order and create a customer invoice;
6. post the invoice;
7. register the payment so the invoice becomes Paid or In Payment, depending on configuration.

The guide emphasises that the price is set in Sales and then flows into the invoice rather than being typed again in Finance.

3.7 Document states matter

The Odoo guide gives a useful general rule: business records move through a lifecycle. A draft is still editable; a confirmed or posted record represents a commitment; a done or paid state represents completion of the real-world event.

Document	Draft-like state	Committed state	Final state
Sales document	Quotation	Sales Order	Locked / fulfilled context
Delivery	Draft	Ready / reserved	Done
Customer invoice	Draft	Posted	Paid / In Payment

If a button is missing, the first diagnostic question should be: *Is the preceding business event complete?*

3.8 Company-wide sales analysis

Tutorial 5 asks students to consider how management could analyse the full relationship with a customer such as Delicious Foods when complete sales data is retained separately in each sales division. In a fragmented environment, company-wide analysis requires collecting data from each division, reconciling customer identities and product names, standardising terms, checking duplicates, and combining the results. Organisational politics can also make divisions reluctant to share detailed performance data.

ERP reduces the technical burden by storing comparable data centrally. It does not automatically remove organisational reluctance, but it makes a unified view possible if governance, authorisation, and management expectations support it.

Worked example

If two divisions sell the same customer under slightly different names—for example “Delicious Foods” and “Delicious Foods Co.”—a manual analysis may treat them as two unrelated customers. Good master-data governance prevents this by maintaining one controlled customer identity that transactions reference.

Try it yourself

Take the old Fitter Snacker process and label every point where information is copied, delayed, estimated, or re-entered. Then map each problem to an ERP mechanism: shared master data, automatic document creation, real-time status, configured pricing, availability checking, or integrated accounting.

Check your understanding

1. Why can a quotation error later become an invoicing problem?
2. What is the difference between a Sales Order and a Delivery in the process model?
3. Why is payment status useful to Sales even though Finance records the payment?

Chapter recap

- The sales order process spans quotation, order, fulfilment, billing, payment, and sometimes returns.
- Fitter Snacker's disconnected systems cause stale credit data, pricing errors, repeated customer questions, missed dates, and manual reconciliation.
- ERP links the commercial, physical, and financial views of the same order.
- Order-to-cash is complete only when fulfilment and settlement are visible across functions.
- Odoo expresses integration through linked records, smart buttons, and document states.

Customer Relationship Management

Chapter at a glance

Before you start: You should know the sales-order process and the difference between a potential sale and a confirmed order.

By the end, you can: explain the purpose of CRM, distinguish operational CRM activities, connect CRM to ERP data, and evaluate how CRM supports acquisition, retention, sales, marketing, and service.

Source basis

This chapter follows Tutorial 4 and the CRM section of Chapter 3 in the textbook. Tutorial 4 uses Luna Electronics and asks students to justify CRM, propose a CRM system, explain its features, and support the proposal with a case study.

4.1 CRM is a relationship process, not just a contact list

Customer Relationship Management (CRM) is a set of processes and systems for managing interactions with prospects and customers across the relationship lifecycle. The goal is to make customer-facing work coordinated, visible, and reusable across the organisation.

A contact database answers “Who is this customer?” A CRM system should also help answer:

- What opportunities are currently active?
- Who is responsible for the next action?
- What has the customer bought before?
- What promises, quotations, complaints, or campaigns has the customer received?
- What is the expected value and probability of the deal?
- What should the salesperson or service representative do next?

Plain idea

CRM tries to give the organisation a “single face to the customer.” The customer should not receive three contradictory prices from Sales, Marketing, and an executive simply because those people do not share interaction history.

4.2 Core CRM activities

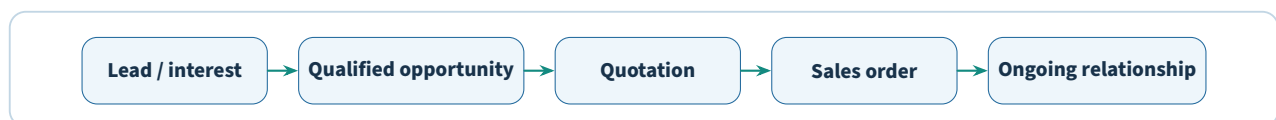
The textbook identifies several recurring CRM capabilities.

CRM activity	Purpose
One-to-one marketing	Tailor offers and communication to a customer's profile or history; supports cross-selling and up-selling.
Sales force automation	Record contacts, assign leads, track opportunities, manage follow-ups, and forecast sales activity.
Campaign management	Organise marketing campaigns and analyse their outcomes.
Marketing content / knowledge	Give staff consistent product and promotional information.
Service and call-centre support	Provide representatives with customer and product history so issues can be resolved without starting from zero.

Modern products may package these features differently, but the business purpose remains similar: capture interactions, coordinate work, and learn from the accumulated customer data.

4.3 Lead, opportunity, quotation, and order are different states

A common beginner mistake is to treat every prospect as a sales order. CRM exists partly because sales work begins before a customer commits.



A **lead** is a potential contact or expression of interest. An **opportunity** is a potential deal being actively pursued. A **quotation** is a commercial offer. A **sales order** is a confirmed customer commitment. After the sale, service and future marketing continue the relationship.

4.4 Why CRM becomes more valuable when integrated with ERP

CRM has customer-facing information; ERP has operational and financial information. When connected, the salesperson does not need to guess whether a customer was delivered, invoiced, or paid.

Useful integrated views include:

- customer purchase history and product mix;
- current quotations and open orders;
- fulfilment and delivery status;
- receivable and payment status;
- profitability or discount patterns;
- service history and recurring complaints.

This supports better decisions such as which customers to prioritise, which products to cross-sell, which accounts need service attention, and whether a discount is commercially sensible.

4.5 The Luna Electronics problem

Tutorial 4 asks how CRM could help Luna Electronics grow its customer base, retain existing customers, and improve customer service and support.

A convincing answer should connect each business problem to a CRM mechanism rather than simply saying “CRM improves customer relationships.”

Worked example

Suppose Luna Electronics loses repeat customers because complaints are handled through personal email inboxes. A CRM process can log the interaction against the customer, assign responsibility, store the response, and make the history visible to the next employee. The business benefit is not the database itself; it is faster, more consistent service and less dependence on one employee’s memory.

Business goal	CRM mechanism	Possible evidence of success
Acquire customers	Lead capture, campaign tracking, opportunity pipeline, follow-up activities.	Lead-to-opportunity and opportunity-to-order conversion.
Retain customers	Interaction history, segmentation, targeted offers, proactive follow-up.	Repeat-purchase rate, churn rate, customer lifetime value trend.
Improve service	Shared case history, knowledge base, ownership and escalation.	Response time, resolution time, repeated-contact rate, satisfaction score.
Improve sales productivity	Pipeline visibility, next activities, opportunity value, automated document hand-off.	Sales cycle length, win rate, forecast accuracy.

4.6 CRM data quality and governance

CRM is only useful if staff record interactions consistently. Common problems include duplicate customer records, opportunities with no next activity, optimistic expected revenue, inconsistent stages, and missing contact details.

A good CRM process therefore defines:

- what counts as a lead and an opportunity;
- when a stage should change;
- who owns the record;
- which fields are mandatory;
- how duplicate customers are prevented or merged;
- what “won” and “lost” mean;
- how lost reasons are recorded and analysed.

Common mistake

A colourful pipeline is not automatically a reliable sales forecast. If users move opportunities between stages inconsistently or enter unrealistic expected revenue, management gets a visually attractive but misleading report.

4.7 CRM in the West End Bicycles Odoo flow

Odoo 19 Community

The supplied Odoo guide creates an opportunity called **Road Bike Order – Seri Iskandar Cycling Club** with expected revenue of RM 5,000. The Sales Manager is assigned as salesperson. The opportunity moves through pipeline stages and then creates the quotation for 5 Road Bikes.

This demonstrates three CRM principles:

1. the deal is owned by a named person;
2. expected revenue exists before the sale is confirmed;
3. the opportunity can create a quotation, preserving the customer relationship history as the deal becomes an operational order.

4.8 How to evaluate a CRM proposal

For the Tutorial 4 proposal, a strong evaluation framework asks six questions:

1. **Fit:** Does the system support the required sales, marketing, and service processes?
2. **Integration:** Can customer, order, delivery, and financial data be connected?
3. **Usability:** Will staff actually record and update interactions?
4. **Reporting:** Can management see conversion, pipeline, campaigns, and service performance?
5. **Governance and security:** Can access, ownership, and customer-data handling be controlled?
6. **Cost and scalability:** Is the system affordable and able to grow with the business?

Try it yourself

For Luna Electronics, choose one customer problem and write a chain in this format: *problem* → *CRM feature* → *changed employee behaviour* → *measurable business outcome*. If one link is vague, the proposal needs more detail.

Check your understanding

1. Why is an opportunity different from a sales order?
2. Give one example of cross-selling and one example of up-selling for Luna Electronics.
3. Why does CRM data quality affect sales forecasting?

Chapter recap

- CRM coordinates customer-facing information before, during, and after a sale.
- Core activities include targeted marketing, sales-force automation, campaign management, and service support.
- Integration with ERP connects relationship history to orders, deliveries, invoices, and payments.
- CRM value should be expressed through changed processes and measurable outcomes, not generic claims.
- Odoo CRM provides the pipeline that turns customer interest into the quotation and sales-order process used later in the course.

Stage 3

Planning, Production, and Supply Chain

Production Planning: From Demand to a Feasible Production Plan

Chapter at a glance

Before you start: You should know that Sales generates demand information and Supply Chain must turn that demand into physical products.

By the end, you can: explain the production-planning hierarchy, distinguish common production approaches, connect forecasts to capacity and inventory decisions, and explain the role of a bill of materials and manufacturing order.

Source basis

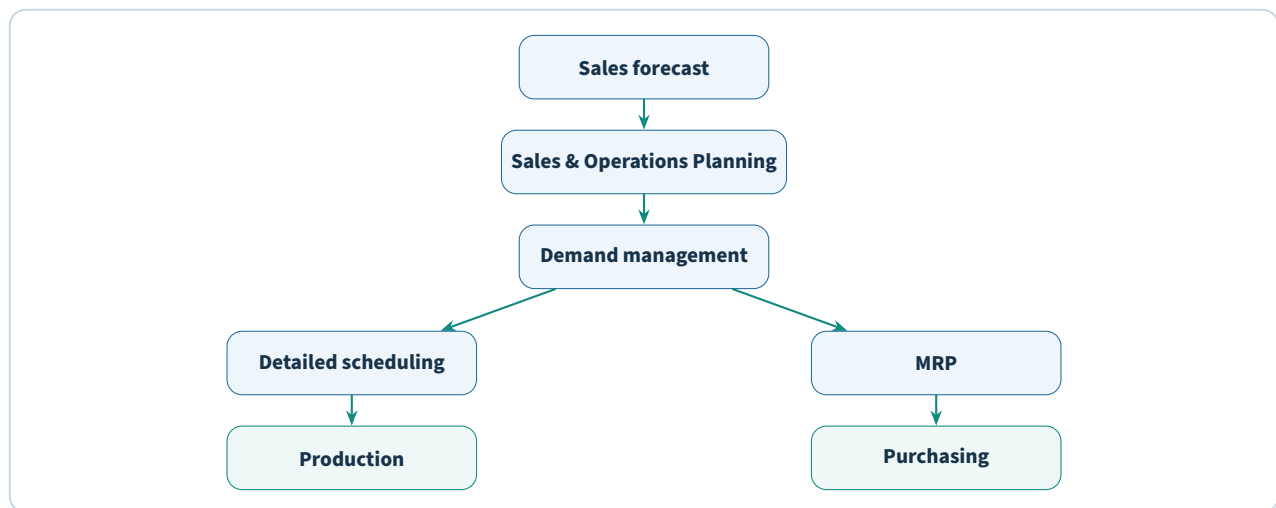
This chapter follows Tutorial 3 (Production Planning) and Chapter 4 of the textbook. The tutorial asks students to select a production-planning approach, calculate a sales forecast, and prepare an S&OP plan. The Odoo build guide provides the Road Bike bill of materials and manufacturing workflow used in the practical.

5.1 What production planning is trying to decide

Production planning coordinates demand, inventory, materials, capacity, and time. The planner must answer several linked questions:

- What do we expect customers to want?
- How much finished-goods inventory do we already have?
- How much should we produce in each period?
- Do we have enough labour and equipment capacity?
- Which raw materials are required, in what quantities, and by what dates?
- When should each product be manufactured?

The textbook organises this into a planning hierarchy.



5.2 Production approaches

The tutorial asks students to choose one production-planning approach for a business venture and understand the alternatives. In introductory ERP courses, the most useful distinction is based on when production happens relative to customer demand.

Approach	Core idea	Typical trade-off
Make-to-stock	Produce finished goods before a specific customer order, based on expected demand.	Fast fulfilment, but inventory risk if demand is wrong.
Make-to-order	Begin production after a customer order is received.	Lower finished-goods inventory, but longer customer lead time.
Engineer/configure-to-order	Design or configure significant product details after the order.	High customisation, but greater planning complexity and longer lead time.

The supplied West End Bicycles Odoo guide explicitly uses **make-to-stock**: Road Bikes are manufactured before the customer order is delivered. That is why raw materials are purchased and 5 bikes are manufactured before the delivery is validated.

5.3 Sales forecasting

A **sales forecast** is an estimate of future demand. It is not a promise and it is not the same as a confirmed order. Forecasts are needed because supply decisions often require lead time.

The textbook demonstrates a simple approach that begins with the previous year's demand, removes the unusual effect of the previous year's promotion, applies expected underlying growth, and then adds the expected effect of this year's promotion.

For month t :

$$\text{Previous-year base}_t = \text{Previous-year sales}_t - \text{Previous promotion}_t$$

$$\text{Base projection}_t = \text{Previous-year base}_t(1 + g)$$

$$\text{Forecast}_t = \text{Base projection}_t + \text{Current promotion}_t$$

where g is the expected growth rate.

Key idea

Forecasting should separate the underlying demand pattern from one-off causes. If last year's promotion temporarily increased sales, simply multiplying the total by a growth rate would treat the temporary promotion as permanent base demand.

5.4 Sales and Operations Planning

Sales and Operations Planning (S&OP) converts the demand forecast into a feasible aggregate production plan. The main balancing relationship is:

$$\text{Ending Inventory}_t = \text{Beginning Inventory}_t + \text{Production}_t - \text{Sales}_t$$

The planner may deliberately produce more than the forecast in one month to build inventory before a peak, or produce less than the forecast while drawing down inventory built earlier.

A feasible plan must also respect capacity:

$$\text{Capacity}_t = \text{Rate per hour} \times \text{Hours per day} \times \text{Working days}_t$$

$$\text{Utilization}_t = \frac{\text{Production Plan}_t}{\text{Capacity}_t} \times 100\%$$

The production plan therefore balances three competing concerns: customer demand, inventory, and capacity.

5.5 Demand management and detailed scheduling

S&OP is intentionally aggregate. The next step, **demand management**, breaks the aggregate plan into individual products and smaller time periods. A monthly production target might become weekly or daily requirements for Road Bikes, Mountain Bikes, and City Bikes.

Detailed scheduling then decides exactly when production runs occur. In repetitive environments, longer runs reduce setup/changeover frequency but can increase finished-goods inventory. Shorter runs can reduce inventory but consume more capacity in setup time.

Plain idea

S&OP decides "how much should the factory produce this month?" Detailed scheduling decides "which product runs on which day, and in what sequence?"

5.6 The bill of materials connects planning to materials

A **bill of materials (BOM)** is the product recipe: a structured list of components and quantities needed to make the finished product. The supplied Road Bike BOM is simple:

Component	Quantity per Road Bike
Aluminium Frame	1
Rubber Tires	2
Brake Set	1
Gear Set	1
Bicycle Chain	1

For 5 Road Bikes, the gross component requirement is therefore 5 frames, 10 tyres, 5 brake sets, 5 gear sets, and 5 chains. MRP later extends this simple multiplication by considering inventory, scheduled receipts, lot sizes, and lead times.

5.7 Manufacturing orders execute the plan

A plan says what should happen. A **manufacturing order (MO)** is the operational instruction to produce a specific quantity. In an integrated ERP system, completing the MO changes inventory: component quantities decrease and finished-goods quantities increase.

Odoo 19 Community

In the West End Bicycles guide:

1. create a Road Bike BOM for one unit;
2. create a Manufacturing Order for 5 Road Bikes;
3. assign the Production Planner as responsible;
4. confirm the MO and ensure components are available;
5. produce and mark the MO Done;
6. verify that component stock fell and Road Bike on-hand stock rose to 5.

This is the physical meaning of integration: production is not only a planning record; it changes the stock records used later by Sales and Delivery.

5.8 Safety stock and the cost of uncertainty

safety stock is extra inventory held to reduce the risk of a stockout when demand or supply is uncertain. More safety stock increases service protection but also increases holding cost and cash tied up in inventory.

Forecast accuracy matters because a less accurate forecast generally requires a larger buffer to achieve the same service level. ERP does not eliminate uncertainty; it improves the information used to manage the uncertainty.

Common mistake

High capacity utilisation is not always the same as good production planning. A factory can keep machines busy by producing goods that customers do not need, creating excess inventory. The objective is an economically sensible balance between demand, capacity, setup cost, and inventory cost.

Try it yourself

Choose one of the business ventures suggested in Tutorial 3—furniture, cookies, printing/personalised products, or your own idea. Decide whether make-to-stock, make-to-order, or a more customised approach fits best. Then explain what would go wrong if you used one of the other approaches instead.

Check your understanding

1. Why can production differ from the sales forecast in a given month?
2. What information does a BOM provide that a sales forecast does not?
3. What stock changes should occur when a Manufacturing Order for 5 Road Bikes is completed?

Chapter recap

- Production planning links forecasts, inventory, capacity, materials, and schedules.
- Forecasting estimates demand; S&OP balances demand with capacity and inventory.
- Demand management and detailed scheduling progressively add product and time detail.
- A BOM defines component requirements; an MO executes production and changes inventory.
- West End Bicycles uses a make-to-stock story: buy components, manufacture bikes, then sell and deliver from finished-goods stock.

Sales Forecasting and Sales & Operations Planning

Chapter at a glance

Before you start: You should understand the production-planning hierarchy and the difference between forecast, production plan, inventory, and capacity.

By the end, you can: calculate a simple promotion-adjusted forecast, construct an S&OP table, interpret capacity utilisation and safety stock, and identify unit or data inconsistencies before using a planning model.

Source basis

This chapter follows Tutorial 6, the supplied Sales Forecast and S&OP spreadsheet, and the forecasting/S&OP sections of Chapter 4 in the textbook. The tutorial uses Footloose Sport Shoes and asks students to automate the calculations with spreadsheet formulas.

6.1 Forecast structure used in the tutorial

The Footloose exercise provides 2022 monthly sales, identifies unusual promotional demand from the previous year, assumes 3% underlying growth, and adds a different promotion plan for the forecast year.

The calculation is best understood as four rows:

1. remove last year's promotion from last year's total to estimate the base;
2. apply the 3% growth assumption to the base;
3. obtain the new base projection;
4. add this year's expected promotion effect.

For a spreadsheet, if the previous-year sales are in row 5, previous promotion in row 6, growth rate in a fixed input cell, and new promotion in row 10, the logic should be formula-driven rather than typed month by month.

6.2 Worked forecast for Footloose Sport Shoes

Using the values in the supplied spreadsheet and the tutorial's 3% growth assumption gives the following results.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Previous year	4000	4000	4500	5000	5000	5500	5500	5750	6000	6000	6200	6300
Previous promo	0	0	0	0	0	200	0	0	0	0	400	400
Base	4000	4000	4500	5000	5000	5300	5500	5750	6000	6000	5800	5900
3% growth	120	120	135	150	150	159	165	172.5	180	180	174	177
Base projection	4120	4120	4635	5150	5150	5459	5665	5922.5	6180	6180	5974	6077
New promo	0	0	0	0	0	0	0	0	0	0	500	400
Forecast	4120	4120	4635	5150	5150	5459	5665	5922.5	6180	6180	6474	6477

If the organisation plans in whole pairs, it must also choose and document a rounding rule. Do not silently round some months and not others.

Worked example

For June, previous-year sales were 5,500 pairs, but 200 pairs were attributed to a promotion. The estimated base is therefore 5,300. Applying 3% growth gives $5,300 \times 1.03 = 5,459$. Since the new promotion is planned only for November and December, the June forecast remains 5,459 pairs.

6.3 S&OP is a balancing problem

Once forecast demand is available, planners decide how much to produce. The inventory balance is:

$$I_t = I_{t-1} + P_t - F_t$$

where I_t is ending inventory, P_t is production, and F_t is forecast demand in period t .

A plan that exactly matches forecast every month keeps inventory unchanged only if beginning inventory is already at the desired level. In practice, production is often shifted between periods because capacity changes with working days and because peak demand may exceed comfortable capacity.

6.4 Capacity and utilisation

Tutorial 6 states that the factory can produce a maximum of 40 shoes per hour, works 8 hours per normal day, and should remain at 99% utilisation or below.

Before calculating, units must be consistent.

Common mistake

The tutorial describes demand in *pairs of shoes* but states a production rate of *40 shoes per hour*. Those are different units. If “40 shoes/hour” is literal, capacity is 20 pairs/hour. If the intended classroom interpretation is 40 pairs/hour, the capacity is twice as large. Confirm the intended unit with the tutor before finalising the S&OP table.

If the intended rate is r pairs/hour, monthly capacity is

$$C_t = r \times 8 \times D_t,$$

where D_t is the number of working days in the month. Utilisation is

$$U_t = \frac{P_t}{C_t} \times 100\%.$$

To enforce the tutorial's ceiling,

$$P_t \leq 0.99C_t.$$

6.5 Safety stock and seasonal peaks

The tutorial says the planner expects higher demand in June and December and wants a safety stock buffer. One sensible strategy is to build inventory in an earlier month when capacity is less constrained, then use that inventory during the peak.

This is exactly why S&OP is more than copying the forecast into the production row. A production plan can intentionally differ from forecast while still satisfying cumulative demand.

6.6 Source inconsistencies to notice before modelling

The course materials contain a few values that should be treated carefully rather than silently reconciled:

- Tutorial 6 asks for the 2023 forecast using 2022 sales, while the supplied spreadsheet headings still say “for year 2022.” The row values nevertheless match the tutorial's previous-year input structure.
- Tutorial 6 states safety stock of 100, but the spreadsheet pre-fills 500 in the December inventory cell of the S&OP table.
- The tutorial uses pairs sold while the stated hourly production rate is in shoes.

Key idea

Planning models are only as reliable as their units and assumptions. Before writing formulas, verify time period, unit of measure, starting inventory, promotion assumptions, capacity rate, and whether the numbers describe individual items, pairs, cases, or product groups.

6.7 A disciplined spreadsheet design

A good planning spreadsheet separates **inputs** from **calculated rows**. The 3% growth rate should be entered once and referenced absolutely. Promotion assumptions should live in their own row. Capacity should be calculated from rate, hours, and working days rather than typed as a number.

A robust design makes assumptions visible so that a tutor or manager can change one input and see the plan recalculate.

6.8 How Odoo contributes to forecasting and planning

The supplied Odoo build guide does not implement a dedicated S&OP workflow. It does, however, create the operational history that planning needs.

Odoo 19 Community

Use Odoo's reporting views to connect the spreadsheet exercise to ERP data:

- **Sales → Reporting**: observe confirmed demand and revenue history.

- **Inventory → Reporting:** inspect on-hand stock and movement history.
- **Manufacturing → Reporting:** review manufacturing orders and output.

The planning spreadsheet is therefore not isolated from ERP; it is an analytical layer that can use historical sales, inventory, and production information generated by the integrated transactions.

6.9 Product-mix planning

The tutorial's S&OP table also allocates production among product types. If total production is P_t and the mix is 50%, 30%, and 20%, then:

$$P_{1,t} = 0.50P_t, \quad P_{2,t} = 0.30P_t, \quad P_{3,t} = 0.20P_t.$$

The percentages must sum to 100%. A product-mix assumption is a simplification: real products may consume different capacities or bottleneck resources. For the tutorial, it is an aggregate allocation rule.

Try it yourself

Build one month of S&OP twice: first under the assumption of 40 *pairs*/hour and then under the literal 40 *shoes*/hour. Compare the resulting capacity and utilisation. This shows why unit discipline matters before any optimisation or planning discussion.

Check your understanding

1. Why should last year's promotion be removed before applying the 3% growth factor?
2. What does a positive ending inventory mean in the S&OP balance equation?
3. Why might production be deliberately higher than forecast in October or November if December demand is expected to peak?

Chapter recap

- The tutorial forecast removes old promotion effects, applies underlying growth, then adds the new promotion.
- S&OP balances forecast demand, production, inventory, working days, and capacity.
- Capacity and utilisation calculations require consistent units.
- Safety stock protects service levels but ties up inventory and cash.
- The supplied Word tutorial and spreadsheet contain a few inconsistent labels/starting values; these should be surfaced and confirmed rather than silently corrected.
- Odoo reporting supplies historical operational data that can feed planning analysis.

Supply Chain Management, MRP, Purchasing, and Inventory

Chapter at a glance

Before you start: You should understand forecasts, production plans, and bills of materials.

By the end, you can: explain supply-chain integration, calculate basic material requirements, distinguish gross from net requirements, connect lead time and lot sizing to purchasing, and evaluate supply-chain performance with operational metrics.

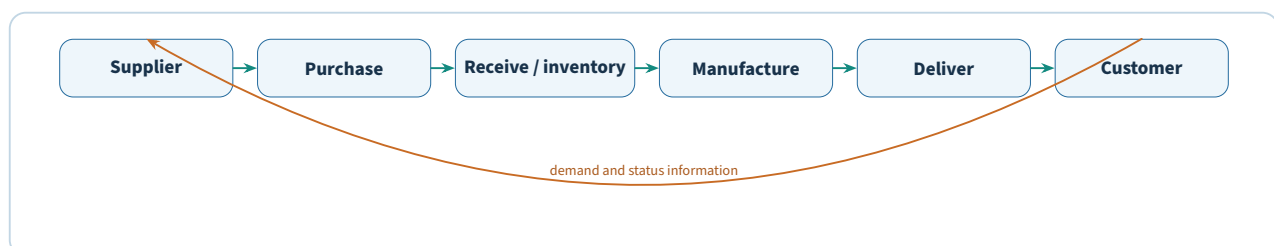
Source basis

This chapter follows Tutorial 7 and Chapter 4 of the textbook. Tutorial 7 asks students to research a manufacturing company that implemented SCM and to compare CRM with SCM. The textbook adds MRP, supplier integration, and supply-chain performance metrics such as cash-to-cash cycle time, fill rate, lead time, and on-time performance.

7.1 What Supply Chain Management coordinates

Supply Chain Management (SCM) coordinates the organisations and activities involved in moving materials, information, and products from suppliers through production and distribution to the final customer.

A traditional view treats each company as an isolated optimiser: the buyer pressures the supplier, the supplier protects its own information, and each party carries inventory to protect itself from uncertainty. An integrated supply-chain view recognises that poor coordination creates cost for everyone.



The physical flow generally moves toward the customer. Information must move both directions: forecasts and orders upstream, availability and shipment status downstream. Money flows back toward suppliers and the producer after transactions are settled.

7.2 From production requirements to material requirements

A sales forecast does not tell Purchasing how many tyres to buy. Production planning first determines what finished products are needed. Then the BOM converts those finished-product quantities into component quantities.

For component j used in products i :

$$\text{Gross Requirement}_j = \sum_i (\text{Planned Product Quantity}_i \times \text{BOM Quantity}_{ij}).$$

For 5 Road Bikes, with 2 tyres per bike:

$$5 \times 2 = 10 \text{ tyres.}$$

That is a **gross requirement**: the amount needed before considering what is already available.

7.3 MRP: gross requirements are not the same as purchase orders

Material Requirements Planning (MRP) determines the quantity and timing of required raw-material and subassembly supply. It considers more than the BOM:

- gross requirements from the production plan;
- current on-hand inventory;
- scheduled receipts from orders already placed;
- safety stock or minimum-stock rules;
- supplier or internal lead times;
- lot-sizing constraints.

A simplified net-requirement idea is:

$$\text{Net Requirement} = \max(0, \text{Gross Requirement} + \text{Desired Ending Stock} - \text{Available Supply}).$$

The exact MRP record is time-phased: availability in one week cannot satisfy a requirement in an earlier week.

7.4 Lead time changes when the order must be released

Lead time is the elapsed time between initiating supply and having the material available for use. If a component is needed in Week 5 and its purchasing lead time is two weeks, the order must be released approximately in Week 3, subject to the system's calendar and receiving assumptions.

MRP therefore answers both *how much* and *when*.

7.5 Lot sizing changes the quantity

Suppliers may sell in fixed packs, pallets, minimum quantities, or economic batch sizes. If the net requirement is 930 units but the supplier only ships pallets of 500, the planned order may need to be 1,000 rather than 930.

Worked example

Suppose West End Bicycles plans 20 Road Bikes. The BOM requires 1 Aluminium Frame and 2 Rubber Tyres per bike. On hand are 7 frames and 12 tyres. Ignoring safety stock and scheduled receipts:

$$\text{Frames net} = 20 - 7 = 13,$$

$$\text{Tyres net} = 40 - 12 = 28.$$

If tyres must be purchased in boxes of 10, the procurement quantity becomes 30 tyres even though the net requirement is 28.

7.6 Purchasing and receiving are separate business events

A purchase order records the organisation's commitment to buy. A receipt records what physically arrived. Those two events may differ in date or quantity.

This distinction matters because:

- Inventory should rise only when goods are actually received;
- supplier performance should be measured against the requested quantity/date;
- Finance should know what was ordered and what was accepted before payment;
- manufacturing availability depends on actual on-hand stock, not merely an open purchase order.

7.7 Supplier integration

The textbook's supply-chain discussion stresses that information sharing and relationships matter. Sharing realistic forecasts and status data can help suppliers prepare capacity and materials. In return, the buyer gains better visibility into lead time and availability.

This collaboration does not mean giving every supplier unrestricted ERP access. It means designing controlled information exchange so the supply chain can coordinate around the same demand and fulfilment signals.

7.8 SCM measures of success

The textbook highlights several metrics.

Metric	Meaning
Cash-to-cash cycle time	Time between paying for inputs and collecting cash from the customer. Shorter cycles generally mean less working capital is tied up.
Total SCM cost	Cost of buying/handling inventory, processing orders, and supporting the information systems used by the supply chain.
Initial fill rate	Percentage of an order supplied in the first shipment.
Initial order lead time	Time required for the supplier to fill the order.
On-time performance	How often the supplier meets the agreed delivery date.

Metrics should be interpreted together. A supplier can have a high on-time percentage but a poor fill rate if it repeatedly delivers on time but short.

7.9 CRM versus SCM

Tutorial 7 asks students to compare CRM and SCM. Both coordinate relationships and information across organisational boundaries, but they focus on different sides of the enterprise.

Aspect	CRM	SCM
Primary relationship	Prospects and customers	Suppliers, production, logistics partners
Primary signal	Customer interest and demand	Material/product requirements and fulfilment
Typical records	Leads, opportunities, contacts, campaigns, service history	Purchase orders, receipts, inventory, production plans, supplier performance
Shared objective	Deliver customer value efficiently by coordinating information rather than optimising one department in isolation.	
Where they meet	Demand captured by CRM/Sales becomes a planning and fulfilment requirement for SCM; SCM status returns to Sales so customers can receive accurate promises.	

7.10 SCM in Odoo 19 Community

Odoo 19 Community

The West End Bicycles guide uses this procure-and-receive sequence:

1. create vendors and raw-material products;
2. link each vendor to the material it supplies and the vendor price;
3. create one RFQ per vendor and assign the Purchasing Officer as Buyer;
4. confirm each RFQ into a Purchase Order;
5. open the linked Receipt and validate it;
6. verify that on-hand raw-material stock rises;
7. use those materials in the Road Bike manufacturing order;
8. later validate the customer delivery, which reduces finished-goods stock.

This makes the supply chain visible as stock movements tied to purchasing and manufacturing records rather than as an isolated warehouse count.

Common mistake

An open Purchase Order is not the same as available inventory. A planner who treats ordered material as already on hand can release production that cannot actually start.

Try it yourself

For the Road Bike BOM, imagine a production plan of 50 bikes, on-hand inventory of 12 frames, 70 tyres, 18 brake sets, 20 gear sets, and 15 chains. Calculate the simplified net requirement for each component. Then add one realistic lot-size rule of your own and recalculate the purchase quantity.

Check your understanding

1. What is the difference between gross and net material requirements?
2. Why must lead time be represented in MRP rather than only the required quantity?
3. How do CRM and SCM interact in a customer-order process?

Chapter recap

- SCM coordinates physical, information, and financial flows across suppliers, production, distribution, and customers.
- MRP converts production requirements into time-phased material requirements using BOMs, inventory, receipts, lead times, and lot sizes.
- Purchase orders and receipts are separate events; only the receipt creates actual stock availability.
- SCM performance should be measured with cycle-time, cost, fill-rate, lead-time, and on-time metrics.
- CRM captures demand; SCM fulfils it; integrated ERP data links the two.

Stage 4

Finance, People, and Process Improvement

Accounting and Finance in an ERP System

Chapter at a glance

Before you start: You should understand the order-to-cash, procure-to-pay, and manufacturing flows.

By the end, you can: explain how operational transactions create financial information, distinguish financial and managerial accounting, describe receivables and credit information, and connect inventory and sales events to management reporting.

Source basis

The tutorial set does not contain a separate Accounting tutorial, but Accounting and Finance appears repeatedly in Tutorial 1, Tutorial 5, and the Odoo build guide. This chapter therefore uses Chapter 5 of the textbook to supply the missing conceptual foundation and maps it to Odoo Invoicing in the practical scenario.

8.1 Why Accounting is embedded in every ERP process

Accounting is not an activity that happens only at the end of the month. Every sale, purchase, receipt, production event, delivery, invoice, payment, payroll run, and expense can have financial consequences.

In an unintegrated environment, accountants must collect and reconcile data from operational systems. In an ERP environment, the objective is to record the business event once and reuse that event to produce the accounting information required by Finance.

Key idea

Operational data and financial data are two views of the same business event. “Five Road Bikes delivered” is an inventory event, a customer-service event, and part of the evidence behind the financial transaction.

8.2 Financial accounting versus managerial accounting

Financial accounting focuses on formal records and statements for external and stewardship purposes. Two core statements introduced in the textbook are the balance sheet and the income statement.

Managerial accounting focuses on internal decision-making: product cost, profitability, budgets, performance, and resource use.

Dimension	Financial accounting	Managerial accounting
Primary users	Owners, creditors, regulators, external stakeholders, management	Internal managers and process owners
Main questions	What is the company's financial position and performance?	Which product, customer, activity, or process is profitable and why?
Typical outputs	Balance sheet, income statement, cash-related reports	Cost reports, budgets, variance analysis, product/customer profitability
Data need	Accurate and controlled transaction records	Detailed operational and financial data with useful dimensions

An ERP system helps because the same central transaction data can support both views.

8.3 Accounts receivable and customer credit

When a company sells on credit, it creates an **account receivable**: an amount the customer owes. Sales needs current receivable and payment information when deciding whether a new order should be accepted under the customer's credit terms.

The Fitter Snacker case shows why stale data is dangerous. A weekly printed credit report may ignore a recent payment or a recent sale. In an integrated ERP, Sales and Finance can use the current customer balance and open transactions instead of waiting for the next report.

8.4 Inventory has both quantity and value

When components are received, Inventory sees a quantity increase. Accounting needs the monetary consequence of that event as well. Similarly, when manufacturing consumes materials and produces finished goods, the system needs to preserve both physical and value information so managers can understand product cost.

The textbook notes that product-cost analysis can use data already maintained in production planning—for example BOM quantities—together with cost information. This reduces the need for accountants to rebuild the material structure manually.

Worked example

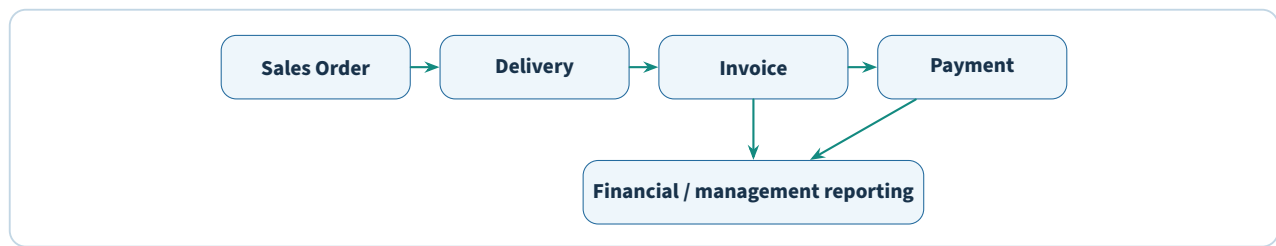
The Road Bike BOM contains one Aluminium Frame at a guide cost of RM 200, two Rubber Tires at RM 40 each, one Brake Set at RM 70, one Gear Set at RM 90, and one Bicycle Chain at RM 30. Ignoring labour and overhead, the direct material guide cost is:

$$200 + 2(40) + 70 + 90 + 30 = \text{RM } 470.$$

If the selling price is RM 1,000, RM 530 is *not* automatically profit because labour, overhead, selling costs, taxes, discounts, and other costs have not yet been considered.

8.5 The accounting document flow

A useful ERP principle is **document flow**: users should be able to trace a transaction through related records rather than search several disconnected systems.



This lineage supports customer service. A salesperson can answer “Has the customer paid?” without asking Finance to search a separate system, assuming the user has appropriate access rights.

8.6 Closing the books and reporting

A company must periodically prepare financial statements and management reports. If source transactions are fragmented and late, accountants spend time finding missing data and posting adjustments. A central ERP database makes it easier to produce current reports because operational events are already stored in a controlled structure.

This does not eliminate accounting judgement or controls. It reduces the mechanical work of gathering inconsistent copies of transactions.

8.7 Controls and authorisation

Integration increases the importance of access control. If one system touches Sales, Inventory, Purchasing, and Finance, users should receive only the permissions required for their roles.

Typical controls include:

- approval limits;
- separation of duties;
- authorised user roles;
- audit trails and document history;
- tolerances for quantity or price differences;
- controlled master-data changes.

Common mistake

A shared database does not mean every employee should see or edit everything. Integration and authorisation are separate design concerns: data can be connected while access remains role-based.

8.8 Finance in the Odoo Community workflow

The supplied guide explicitly states that the course uses **Invoicing** in Odoo 19 Community rather than the broader Enterprise Accounting feature set.

Odoo 19 Community

For the Road Bike order:

1. the Sales Order contains 5 Road Bikes at RM 1,000 each;
2. **Create Invoice** generates a draft invoice from the Sales Order;
3. posting the invoice changes it from Draft to Posted;

4. **Register Payment** records the customer payment in Bank or Cash using the manual method for the demo;
5. the invoice becomes Paid or In Payment depending on configuration.

The important information-flow lesson is that product, quantity, customer, and price come from the linked sales transaction instead of being re-entered.

8.9 Management reporting

The Odoo guide recommends Graph and Pivot views after the end-to-end transaction is complete. The point is not merely to create a chart: the report is generated from the transaction history built by Sales, Inventory, Manufacturing, and Invoicing.

Useful management questions include:

- How much was sold and invoiced?
- Which products generated revenue?
- Which invoices remain unpaid?
- What inventory remains on hand?
- What was manufactured during the period?
- Which customer opportunities are still open?

Try it yourself

Trace one West End Bicycles transaction from Sales Order to payment. At each step write two columns: “operational fact” and “financial/management meaning.” For example, “5 bikes delivered” is the operational fact; “customer order fulfilled and inventory reduced” is one management meaning.

Check your understanding

1. Why does Sales need current Accounts Receivable information?
2. What is the difference between direct material cost and profit?
3. Why is document flow useful for both accountants and customer-facing staff?

Chapter recap

- Accounting in ERP is integrated with operational events rather than reconstructed from periodic transfers.
- Financial accounting reports overall financial position and performance; managerial accounting supports internal decisions and profitability analysis.
- Inventory has both physical quantity and financial value.
- Up-to-date receivable and payment data improves credit and customer decisions.
- Odoo’s course workflow demonstrates sales-to-invoice-to-payment information flow in Community Edition.

Human Resources Processes and Recruitment

Chapter at a glance

Before you start: You should understand master data, process ownership, and why cross-functional information must be current.

By the end, you can: describe an end-to-end recruitment process, identify information problems that harm candidate experience, explain how ERP-supported HR records improve coordination, and distinguish an employee record from an operational system user.

Source basis

This chapter follows Tutorial 8 and Chapter 6 of the textbook. The tutorial uses the ABC Company recruitment scenario and asks students to identify information-system problems and draw both a flowchart and a swimlane diagram. The Odoo guide adds department, employee, manager, and user-role setup for West End Bicycles.

9.1 Human Resources is a business process partner

HR is often introduced through administrative tasks, but its information flows support every functional area. Departments need people with the right skills; HR needs accurate vacancy, job, salary, and reporting information from those departments.

Typical HR processes include recruiting, hiring, onboarding, training, compensation administration, benefits, performance management, compliance, and workforce planning.

9.2 Recruitment as an end-to-end process

The ABC Company scenario in Tutorial 8 can be expressed as a sequence of business events:

1. a department identifies a staffing need;
2. the manager submits a hiring request and position requirements;
3. HR performs job analysis and prepares the posting;
4. the vacancy is advertised;
5. applications are received and screened;
6. shortlisted candidates are interviewed;
7. top candidates complete a skills assessment;
8. references and background checks are completed;
9. stakeholders review the evidence and select a candidate;
10. HR issues the offer with salary, benefits, and start date.

The textbook's Fitter Snacker example shows how a paper-based version of this process can fail: forms may be inconsistent or lost, applications can be misfiled, reviews happen sequentially because paper copies must circulate, interview scheduling is slow, and promising candidates can accept another offer while the company is still coordinating internally.

9.3 Candidate experience is an information-flow problem

Tutorial 8 asks which problems could create a negative candidate impression. Useful categories include:

Problem	What the candidate experiences	Information-system response
Slow scheduling	Long silence between stages	Shared calendars, workflow tasks, status tracking, reminders
Lost or duplicated documents	Candidate asked to resend information	Central applicant record and document control
Inconsistent requirements	Different interviewers describe the job differently	Controlled job description and competency profile
Missing feedback	Repeated delays while decision-makers chase comments	Structured evaluation forms and completion status
Poor communication	Candidate does not know whether the application is active	Standardised stage/status communication
Incorrect offer details	Salary, role, or start date conflicts	Approved master data and controlled offer workflow

Key idea

A candidate experiences the organisation as one process. They do not care that scheduling belongs to HR, technical evaluation belongs to the hiring manager, and approval belongs to senior management. ERP/process integration should hide that internal fragmentation from the candidate.

9.4 Job information should be treated as controlled data

A job vacancy should not be a hurried free-text note. The organisation benefits from maintaining structured information such as:

- job title and department;
- reporting line;
- responsibilities;
- required and preferred skills;
- qualifications and experience;
- employment type;
- location;
- compensation range or grade where appropriate;
- required approvals;
- planned start date.

This makes candidate screening more consistent and later supports onboarding, training, performance evaluation, and workforce planning.

9.5 The “dream job” activity as a data-design exercise

Tutorial 8 asks students to choose a desired role and identify the information needed to judge candidates. The point is not merely to list attractive skills. A good answer distinguishes:

Evidence Degree, certifications, portfolio, previous roles, project outcomes, technical tests, references.

Competencies Technical knowledge, communication, problem solving, teamwork, leadership, domain knowledge.

Constraints Work authorisation, location, availability, salary expectations, required schedule.

Selection criteria Which attributes are essential, which are preferred, and how each is evaluated.

That structure turns recruitment into a repeatable business process rather than an informal impression.

9.6 Employee master data and organisational structure

Once someone is hired, HR records become master data reused throughout the organisation: employee identity, department, manager, role, work contact information, and possibly compensation or training information subject to authorisation.

Odoo 19 Community

The West End Bicycles Odoo guide creates four departments:

- Sales and Marketing;
- Accounting and Finance;
- Supply Chain;
- Human Resources.

It then creates employees such as Sales Manager, Purchasing Officer, Production Planner, Accountant, and HR Officer, assigns managers, and builds a simple organisation chart.

9.7 Employee record versus system user

The Odoo guide makes an important systems distinction: an **Employee** record and a **User** login are different objects.

An employee record describes a person in the organisation. A user account gives the person authenticated access and permissions in Odoo. Operational document fields such as Salesperson, Buyer, and Responsible may reference users because the system needs to know who can own or act on a transaction.

The guide therefore creates user accounts for selected operational roles and links them back to the corresponding employee records.

Role	Operational ownership	Example in the Odoo guide
Sales Manager	Salesperson / team leadership	CRM opportunity, quotation, Sales Order
Purchasing Officer	Buyer	RFQ and Purchase Order
Production Planner	Responsible	Manufacturing Order
Accountant	Finance activity	Customer payment and invoice follow-up

9.8 HR data and access control

HR information can be sensitive. ERP integration must therefore combine availability with authorisation. A production manager may need an employee's skills or availability without needing access to private compensation or personal information.

This is a good example of why “one shared database” does not mean “one unrestricted database.”

Common mistake

Do not solve HR integration by giving everyone broad access. A well-designed ERP system shares the right data with the right role while preserving confidentiality and accountability.

9.9 From recruitment to the rest of ERP

After hiring, HR information supports other processes:

- training records determine whether staff are qualified for specialised work;
- department and manager relationships support approvals and responsibility;
- workforce forecasts support production and sales plans;
- payroll and benefit expenses feed financial reporting;
- access rights connect employee responsibilities to system permissions.

Try it yourself

Choose one role from the Tutorial 8 “dream job” activity. Create a compact selection matrix with five criteria, the evidence used to judge each criterion, and an importance rating. Then identify which fields belong in long-term employee master data and which belong only in the recruitment transaction.

Check your understanding

1. Name three ways a paper-based recruitment process can lose a strong candidate.
2. Why are an Odoo Employee and an Odoo User conceptually different?
3. Give one example of HR information needed by another functional area and one example of information HR needs from that area.

Chapter recap

- Recruitment is a cross-functional process involving the hiring department, HR, candidates, assessors, and approvers.

- Slow or inconsistent information flow directly affects candidate experience and hiring quality.
- Structured job and candidate data makes recruitment more repeatable and auditable.
- Employee master data supports organisation structure, training, finance, and access management.
- In Odoo, HR employee records and login users serve different purposes and must be linked when operational ownership is required.

Process Modeling, Process Improvement, and ERP Implementation

Chapter at a glance

Before you start: You should understand end-to-end business processes and the implementation challenges introduced earlier.

By the end, you can: draw basic flowcharts and swimlane diagrams, use process models to expose hand-offs and delays, distinguish current-state from future-state design, and explain the main phases and change-management needs of an ERP implementation.

Source basis

This chapter follows the process-modelling part of Tutorial 8 and Chapter 7 of the textbook. It also supports Tutorial 2 by expanding the implementation and change-management concepts behind ERP benefits and challenges.

10.1 Why model a process before changing it

A process model turns a verbal description into a visible sequence of events, decisions, responsibilities, and hand-offs. This is useful because many ERP problems are not obvious until the process is drawn.

A model can reveal:

- repeated data entry;
- unnecessary approvals;
- unclear ownership;
- long queues or waiting points;
- hand-offs between departments;
- missing decision rules;
- steps that exist only because old systems were disconnected.

10.2 Basic flowchart symbols

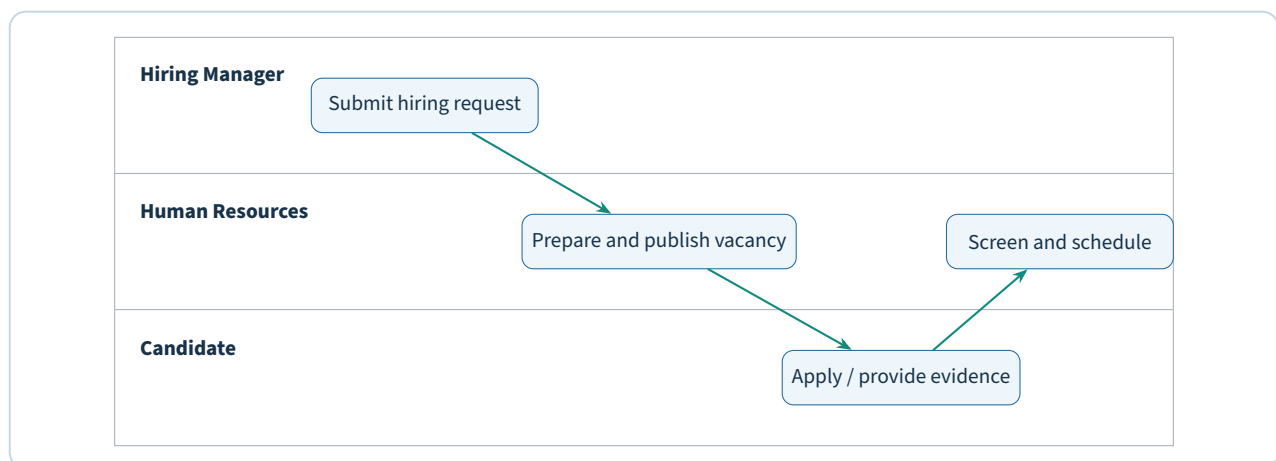
The textbook introduces flowcharts as a simple way to represent a process from beginning to end. For this course, four symbol roles are enough for most tutorial work.

Symbol role	Typical shape	Meaning
Start / end	Rounded terminator	Boundary of the process.
Activity	Rectangle	Work performed, such as “Screen resumes.”
Decision	Diamond	Branch based on a condition, such as “Candidate suitable?”
Flow	Arrow	Direction and sequence of the process.

The most important rule is semantic clarity: every activity should use a verb and describe one meaningful piece of work.

10.3 Swimlane diagrams add responsibility

A **swimlane** or deployment flowchart adds lanes for actors, departments, or roles. This makes cross-functional hand-offs explicit.



A normal flowchart answers “What happens next?” A swimlane also answers “Who is responsible when it happens?”

10.4 Current-state and future-state models

For ERP implementation, teams often need two versions:

As-is model How the process operates today, including manual workarounds, delays, and system boundaries.

To-be model How the process should operate after redesign and ERP configuration.

The goal is not simply to redraw the same process with Odoo or another ERP logo. The to-be process should remove waste where possible.

Common mistake

If an old process requires a clerk to retype a sales order into Finance because the systems are separate, reproducing that same step after ERP implementation defeats the purpose of integration.

10.5 Process improvement before automation

A useful improvement sequence is:

1. identify the customer or internal user and the process outcome they value;
2. map the current process;
3. measure delay, rework, error, cost, and hand-offs;
4. challenge steps that do not create value or control risk;
5. redesign the process;
6. configure the ERP workflow to support the redesigned process;
7. measure the outcome after implementation.

This prevents *automating waste*.

10.6 ERP implementation as a staged project

The textbook presents a five-phase implementation roadmap associated with SAP Solution Manager. The names are useful even when another ERP product is used because they describe general project logic.



10.6.1 Project Preparation

Define scope, objectives, governance, team, technical landscape, high-level schedule, and success measures.

10.6.2 Business Blueprint

Document business requirements and the intended process design. Process mapping is critical here because the team must agree how the company will operate in the ERP system.

10.6.3 Realization

Configure the ERP system, build required interfaces or controlled extensions, prepare data, and develop supporting materials.

10.6.4 Final Preparation

Test critical processes and transaction volume, prepare support, migrate data, train users, and set the go-live date.

10.6.5 Go Live and Support

Move into production, stabilise the system, resolve issues, and continue improvement.

10.7 Scope creep

Scope creep is the unplanned expansion of project goals. It increases time, cost, complexity, and risk. The most dangerous form occurs when the project adds new features faster than it completes testing, training, and process readiness.

A practical defence is a change-control rule: every new requirement should state the business value, impact, cost, owner, and whether it is essential for go-live or can wait for a later phase.

10.8 Organisational change management

ERP implementation changes roles, responsibilities, information visibility, and sometimes staffing. The textbook emphasises that people often resist being changed, especially when they feel the project is imposed on them.

Good **organisational change management (OCM)** includes:

- involving process owners and end users in design;
- communicating why the process is changing;
- explaining role and responsibility changes;
- training for both system use and new business procedures;
- providing support during the transition;
- measuring adoption rather than assuming it.

Key idea

ERP implementation succeeds when the organisation changes how work is done, not when the technical team merely installs software. Process ownership, training, testing, data quality, and executive sponsorship are therefore core implementation work.

10.9 The ABC recruitment process as a modelling exercise

Tutorial 8 is ideal for a swimlane because responsibility moves among the hiring manager, HR, candidate, interviewers, and approval stakeholders. The model should expose waiting points such as scheduling interviews, collecting feedback, and conducting checks.

A future-state information system could reduce waiting by making status, documents, interview feedback, and ownership visible in one workflow.

Try it yourself

Draw the ABC recruitment process twice: first as a simple flowchart and then as a swimlane. In the swimlane version, mark every arrow that crosses a lane. Those crossings are the hand-offs most likely to suffer from delay or missing information.

Check your understanding

1. What extra information does a swimlane provide compared with a basic flowchart?
2. Why is an as-is process model necessary before configuring an ERP workflow?
3. What is scope creep and why can it indirectly reduce testing or training quality?

Chapter recap

- Process models expose sequence, decisions, responsibilities, hand-offs, and waste.
- Swimlanes are especially useful for cross-functional ERP processes.
- ERP projects should redesign weak processes rather than copy them exactly.

- Implementation proceeds from scope and blueprint through configuration, preparation, go-live, and support.
- Change management, training, data migration, and process ownership are essential because ERP changes how people work.

Stage 5

Integrated Odoo Practicals

PRACTICAL MILESTONE

Lab 1 of 8

Business Process and Odoo Environment Setup

West End Bicycles: map the organisation, then build the ERP foundation

Coverage. Tutorial 1 + Chapters 1–2.

Core system. Odoo 19 Community Edition.

Goal. Connect functional-area theory to a real ERP structure by creating the company, required apps, departments, employees, and operational users.

How the practical connects to the concept

The tutorial asks students to identify business functions, discuss cross-functional Sales teams, describe information flow between Sales and Finance, and draw a purchasing process. The Odoo build makes those abstract boxes concrete: departments become records, employees become owners, and later transactions will be assigned to them.

Part A — Odoo 19 orientation.

Use the navigation conventions from the supplied guide:

- main app grid at the top-left;
- app-specific menu bar across the top;
- breadcrumb trail for moving back through linked records;
- smart buttons for jumping to related documents;
- new records through **New**; existing records are directly editable;
- cloud/save indicator for unsaved changes, with auto-save when navigating away.

Common mistake

Older Odoo tutorials may refer to a separate Edit button or the old “Storable Product” type. The supplied Odoo 19 guide states that physical stock items use **Product Type = Goods** plus **Track Inventory**.

Part B — Create or access the database.

For a local Community installation, use the Database Manager route described in the guide and create a database such as `west_end_bicycles`. Use Malaysia as country so the company context matches the supplied build. If the class deployment is already provisioned, use the assigned database instead.

Part C — Install the course apps.

Install or confirm these eight apps:

App	Role in the course
Contacts	Shared customer and vendor records.
CRM	Opportunities and pipeline.
Employees	Departments, people, managers, and organisation structure.
Inventory	Receipts, deliveries, and on-hand stock.
Invoicing	Customer invoices and payments in the Community workflow.
Manufacturing	BOMs and Manufacturing Orders.
Purchase	RFQs, Purchase Orders, and vendor receipts.
Sales	Quotations and Sales Orders.

Part D — Create the company and organisation.

Set the company to **West End Bicycles**, Ipoh, Perak, Malaysia. Then create four departments:

- Sales and Marketing;
- Accounting and Finance;
- Supply Chain;
- Human Resources.

Add the sample employees from the supplied guide and set department heads. At minimum, ensure Sales Manager, Purchasing Officer, Production Planner, Accountant/Finance role, and HR Officer are present.

Part E — Create operational users.

The guide distinguishes HR employee records from login users. Create and link user accounts for the operational roles needed later so they can appear in fields such as Salesperson, Buyer, and Responsible.

Part F — Answer the Tutorial 1 concept questions.

Using West End Bicycles, prepare:

1. at least three business functions for each of the four functional areas;
2. a reasoned choice about whether the customer-order Sales team should include only Sales/Marketing staff or cross-functional members;
3. a description of information flowing between Marketing/Sales and Accounting/Finance;
4. a business-process diagram for ordering 1,000 bicycles from a supplier.

Try it yourself

For Question 3, do not write only “Sales sends order data to Finance.” Include the reverse flow too: customer credit/payment status and invoice/payment information must return to Sales so the customer can receive an accurate answer.

Evidence to capture.

- installed-apps screen;
- completed company profile;
- departments view with managers;
- one organisation-chart view;

- user list showing the operational logins.

Lab deliverable

Submit the Tutorial 1 answers plus screenshots proving the Odoo foundation is ready. Your process diagram should show at least the request, supplier/order step, receipt, inventory update, and financial/payment consequence rather than a single departmental box.

PRACTICAL MILESTONE

Lab 2 of 8

ERP Benefits, Challenges, and Modular Integration

Teach one ERP benefit and one ERP challenge using the system you just built

Coverage. Tutorial 2 + Chapter 2.

Core system. Odoo 19 Community app structure.

Goal. Move from generic statements such as “ERP improves efficiency” to evidence-based explanations of how integration creates benefits and why implementation is difficult.

How the practical connects to the concept

Tutorial 2 uses Team Benefit and Team Challenge. Odoo is used here as a concrete reference architecture: each team must point to at least one app-to-app connection or setup choice that demonstrates the selected benefit/challenge.

Part A — Team Benefit.

Choose one ERP benefit such as shared data, faster process flow, management visibility, reduced duplicate entry, better inventory accuracy, or integrated reporting. Explain:

1. the old fragmented problem;
2. the ERP mechanism that changes it;
3. the Odoo record or connection that demonstrates the mechanism;
4. one metric that could show the benefit.

Example structure: “A Sales Order creates a linked Delivery and can create an Invoice. This reduces repeated entry of product, quantity, customer, and price. Measure the percentage of invoices requiring manual correction.”

Part B — Team Challenge.

Choose one challenge such as data migration, training, change resistance, scope creep, poor process design, weak sponsorship, role/permission design, or excessive customisation. Explain:

1. why it is difficult;
2. which users/processes are affected;
3. what could happen if it is ignored;
4. how the implementation team should reduce the risk.

Part C — Inspect modularity in Odoo.

Open the app grid and identify which app owns each of these records: customer, opportunity, quotation, purchase order, receipt, BOM, manufacturing order, delivery, invoice, employee. Then draw arrows showing which records create or feed the next record.

Key idea

The point is to see modularity without silos. The records live in different apps because the workflows differ, but they remain connected through shared master data and document relationships.

Part D — Teach the other subgroup.

Follow Tutorial 2: explain your selected topic, listen to the other subgroup, take notes, and write the required paragraph demonstrating what you learned.

Lab deliverable

Submit the paragraph required by Tutorial 2. Keep your research source details separately as required by the tutorial. In class, be ready to demonstrate one Odoo screen or linked-record example that makes your benefit/challenge concrete.

PRACTICAL MILESTONE

Lab 3 of 8

Production Planning and Manufacturing in Odoo

Turn a production plan into products, a BOM, and a completed Manufacturing Order

Coverage. Tutorial 3 + Chapter 5.

Core system. Odoo Products, Purchase, Inventory, and Manufacturing.

Goal. Connect production-planning concepts to the master data and transactions required to manufacture Road Bikes.

How the practical connects to the concept

Tutorial 3 asks for a production approach, forecast, and S&OP plan. Odoo does not replace those planning calculations in this course. Instead, it demonstrates how an approved plan becomes executable master data and manufacturing transactions.

Part A — Choose the production approach.

For your Tutorial 3 business venture, justify make-to-stock, make-to-order, or a more customised approach. For West End Bicycles, use the supplied guide's make-to-stock story: build finished bikes before the customer delivery.

Part B — Create finished products.

Create Road Bike and, if required by the class, the additional sample finished products. For Road Bike:

- Product Type = Goods;
- Track Inventory = enabled;
- Can be Sold = enabled;
- Sales Price = RM 1,000.

Part C — Create raw materials.

Create Aluminium Frame, Rubber Tires, Brake Set, Gear Set, and Bicycle Chain as Goods with Track Inventory enabled and Can be Purchased enabled. Use the guide costs as reference values.

Part D — Build the Road Bike BOM.

Use **Manufacturing → Products → Bills of Materials**. Create a BOM for 1 Road Bike with:

- 1 Aluminium Frame;
- 2 Rubber Tires;
- 1 Brake Set;
- 1 Gear Set;
- 1 Bicycle Chain.

Before manufacturing, verify that components have actually been received into stock. The SCM lab explains purchasing in depth; for this lab, use stock created through the supplied purchase/receipt steps or the instructor's prepared database.

Part E — Manufacture 5 Road Bikes.

Create a Manufacturing Order for quantity 5, assign the Production Planner as Responsible, confirm the order, check component availability, and complete production using the Odoo 19 flow described in the supplied guide. Mark the MO Done.

Part F — Verify the inventory transformation.

Before completion, record component on-hand quantities. After completion, verify:

- frame stock decreased by 5;
- tyre stock decreased by 10;
- each other component decreased by 5;
- Road Bike stock increased by 5.

This is the ERP evidence that the manufacturing event changed the shared inventory data.

Part G — Complete the Tutorial 3 planning work.

Prepare the sales forecast and S&OP table required by the tutorial, including the selected product mix and the 90–95% utilisation requirement specified in that tutorial. Keep the planning units consistent.

Lab deliverable

Submit the Tutorial 3 presentation content plus screenshots of the Road Bike product, BOM, completed Manufacturing Order, and the before/after stock evidence. Your explanation must connect the BOM and MO to the production plan rather than treating them as unrelated Odoo forms.

PRACTICAL MILESTONE

Lab 4 of 8

Customer Relationship Management in Odoo

Build a customer opportunity and convert relationship data into a quotation

Coverage. Tutorial 4 + Chapter 4.

Core system. Odoo CRM + Sales.

Goal. Demonstrate that CRM manages the relationship before the sales order exists and preserves that context when the deal becomes a quotation.

How the practical connects to the concept

Tutorial 4 asks for a CRM proposal for Luna Electronics. The Odoo exercise gives you a working example of pipeline, ownership, expected revenue, stages, and conversion to quotation. Use that hands-on experience when explaining what CRM features actually do.

Part A — Create the customer.

Create **Seri Iskandar Cycling Club** as a company contact with the location and contact details from the supplied guide.

Part B — Create the opportunity.

Open **CRM** and create:

- Opportunity: Road Bike Order – Seri Iskandar Cycling Club;
- Customer: Seri Iskandar Cycling Club;
- Expected Revenue: RM 5,000;
- Salesperson: Sales Manager.

Explain why RM 5,000 is an expected amount rather than confirmed sales revenue at this stage.

Part C — Move the deal through the pipeline.

Move the opportunity from New to Qualified and then to Proposition as appropriate. Add an internal note such as “Customer requested 5 Road Bikes.” After the sales order is later confirmed, mark the opportunity Won.

Part D — Create the quotation from the opportunity.

Use the opportunity’s quotation action. Confirm the customer and salesperson, add 5 Road Bikes, and check RM 1,000 unit price. Save the quotation but do not rush past the CRM meaning: the customer, owner, expected revenue, and sales document are now linked.

Part E — Use the experience in the Luna Electronics proposal.

For each proposal section, connect the claim to a CRM mechanism you saw:

- customer acquisition: leads/opportunities and pipeline;
- retention: shared interaction history and follow-up;
- service: a common customer record and visible prior interactions;
- sales productivity: ownership, stages, expected revenue, and conversion to quotation;
- management visibility: pipeline and reporting views.

Common mistake

Do not present expected revenue as money already earned. Pipeline value is a forecast of potential business and depends on stage quality, probability assumptions, and user discipline.

Lab deliverable

Prepare the 10-slide CRM proposal structure required by Tutorial 4. Include Odoo screenshots of the opportunity/pipeline and quotation as practical evidence of CRM features. The proposal still needs one real-company case study and should explain outcomes, not merely list software functions.

PRACTICAL MILESTONE

Lab 5 of 8

Sales Order Process: Order-to-Cash in Odoo

Complete the Road Bike order from quotation to payment

Coverage. Tutorial 5 + Chapters 3 and 8.

Core system. Odoo CRM, Sales, Inventory, and Invoicing.

Goal. Demonstrate a clean order-to-cash document flow and contrast it with the problems in Fitter Snacker's old process.

How the practical connects to the concept

Tutorial 5 focuses on the problems of an unintegrated sales process. This lab deliberately makes the same information move through linked Odoo records so you can point to what ERP integration changes.

Part A — Confirm the quotation.

Open the quotation for Seri Iskandar Cycling Club with 5 Road Bikes at RM 1,000 each. Confirm it. The status becomes Sales Order and linked fulfilment/billing actions become available.

Part B — Deliver the bikes.

Open the Delivery smart button from the Sales Order. Check availability, confirm that 5 Road Bikes are reserved, and Validate the delivery. The delivery status becomes Done and Road Bike on-hand stock should fall by 5.

If Odoo cannot reserve the bikes, diagnose the upstream process: the Manufacturing Order may not be Done or finished stock may be insufficient.

Part C — Create and post the invoice.

Return to the Sales Order and choose Create Invoice. Create the regular invoice, review the Road Bike line, and post/confirm the invoice so it moves from Draft to Posted.

Observe that customer, product, quantity, and price were carried forward from Sales.

Part D — Register payment.

From the posted invoice, choose Register Payment. Use Bank or Cash, Manual method, full amount, and the current date. Create the payment. The invoice should become Paid or In Payment depending on configuration.

Part E — Compare old process versus ERP process.

Use Tutorial 5 Question 1 to compare the salesperson experience:

Issue	Old Fitter Snacker process	Integrated Odoo process
Price and terms	Handwritten/faxed quotation, possible arithmetic or discount errors	Product/order lines preserve the commercial terms in the linked document flow
Availability	Warehouse estimate based on incomplete visibility	Delivery reservation uses current stock state
Fulfilment status	Sales may need to call warehouse	Sales Order links directly to Delivery status
Invoice data	Periodic transfer / re-entry to accounting	Invoice created from the Sales Order
Payment status	Delayed visibility	Payment status shown on linked invoice

Part F — Answer Tutorial 5 Questions 3 and 5.

For the company-wide Delicious Foods analysis, describe the data consolidation and governance problems created by divisional systems. For the CIO memo activity, focus on the information-flow problem and explain how integrated order, fulfilment, invoice, and payment records change management visibility.

Lab deliverable

Submit the Tutorial 5 work plus screenshots of the confirmed Sales Order, Done Delivery, Posted invoice, and Paid/In Payment invoice. Be able to narrate which department owns each step and which information is reused rather than retyped.

PRACTICAL MILESTONE

Lab 6 of 8

Sales Forecasting and S&OP with ERP Data

Use the supplied planning spreadsheet and connect its inputs back to Odoo transactions

Coverage. Tutorial 6 + Chapter 6.

Core systems. Supplied Excel spreadsheet + Odoo Sales/Inventory/Manufacturing reporting.

Goal. Build formula-driven forecast and S&OP logic, then explain where an integrated ERP would supply the historical data used by the model.

How the practical connects to the concept

The tutorial explicitly requires the supplied spreadsheet. Odoo remains part of the lab by providing the operational context: Sales generates history, Inventory provides starting stock, and Manufacturing provides actual production. The spreadsheet performs the aggregate planning calculation.

The spreadsheet was not included with the course materials. Rather than leave the lab uncompletable, Lab06/sop_planner.py reproduces the same calculation with every formula written out:

```
py sop_planner.py
py sop_planner.py --safety-stock 500 --unit pairs
```

Build the spreadsheet as the tutorial requires — that is the deliverable. Use the script to *check* it. If your sheet and the script disagree, one of you has a formula wrong, and finding out which is worth more than either answer.

Part A — Complete the simple sales forecast.

Use the provided previous-year values. For each month:

1. Previous Year Base = Previous Year Sales — Previous Promotion;
2. Growth = Previous Year Base $\times 3\%$;
3. Base Projection = Previous Year Base + Growth;
4. Forecast = Base Projection + Current Promotion.

Current promotion assumptions: +500 pairs in November and +400 pairs in December.

Use formulas. Do not type the calculated rows manually.

Part B — Resolve units before capacity calculation.

The tutorial states demand in pairs but capacity as 40 shoes/hour. Record which interpretation your tutor intends. Your formulas must use one consistent unit throughout.

Part C — Build the S&OP plan.

Calculate capacity from production rate, 8 hours/day, and monthly working days. Keep utilisation at or below 99% as required in Tutorial 6. Use the inventory balance:

$$I_t = I_{t-1} + P_t - F_t.$$

Plan around the expected festive-demand pressure in June and December rather than merely copying forecast into production.

Part D — Check the source discrepancy.

Tutorial 6 states safety stock = 100, while the supplied spreadsheet contains 500 in the initial December inventory cell. Do not hide the conflict. Confirm which value should be used for assessment and state it in your sheet or submission notes.

Part E — Inspect related Odoo reports.

After completing earlier labs, open:

- **Sales → Reporting;**
- **Inventory → Reporting;**
- **Manufacturing → Reporting.**

For each report, identify which S&OP input or validation measure it could provide in a larger implementation: sales history, starting inventory, production output, or actual-versus-plan comparison.

Key idea

Forecasting is not “outside ERP.” Even when the planning calculation is performed in a spreadsheet, the reliability of the model depends on integrated historical sales, inventory, and production data.

Lab deliverable

Submit the screenshot of the completed forecast and S&OP tables required by Tutorial 6. Include formulas, clearly state the unit interpretation and starting-inventory assumption, and add a short note naming the Odoo reports that would supply/validate the underlying operational data.

PRACTICAL MILESTONE

Lab 7 of 8

Supply Chain Management: Procure, Receive, and Measure

Build the upstream supply chain for Road Bikes and connect it to SCM performance

Coverage. Tutorial 7 + Chapter 7.

Core system. Odoo Purchase + Inventory + Manufacturing.

Goal. Demonstrate supplier master data, purchase orders, receipts, inventory availability, and the link between supplier performance and production readiness.

How the practical connects to the concept

Tutorial 7 asks students to research an SCM implementation and compare CRM with SCM. The Odoo work gives a concrete supply-chain process you can analyse using metrics such as lead time, fill rate, and on-time performance.

Part A — Create vendors and vendor-product links.

Create the supplied vendors and link them to raw materials on the product Purchase tab:

- Ipoh Metal Supply → Aluminium Frame;
- RubberWheel Co. → Rubber Tires;
- BikeParts Supplier → Brake Set, Gear Set, Bicycle Chain;
- CarbonTech Components → Carbon Fiber Frame if used in your extended catalogue.

Set vendor prices from the guide and record a delivery lead time if the field is present.

Part B — Raise RFQs and confirm Purchase Orders.

Use one RFQ per vendor. Assign the Purchasing Officer as Buyer. Use the guide's example quantities or quantities approved by the tutor. Confirm each RFQ so the document becomes a Purchase Order and a Receipt smart button appears.

Part C — Receive materials.

Open the linked Receipt, check quantities, and Validate. Repeat for each supplier. Verify on-hand stock for one component before and after validation.

Part D — Explain why receipt is not the same as order.

Record one sentence for each:

- what the Purchase Order proves;
- what the Receipt proves;
- which event makes components usable by Manufacturing;
- which event should supplier on-time/fill-rate analysis compare against.

Part E — Build a supplier scorecard concept.

For each supplier, define:

- initial fill rate;
- order lead time;
- on-time performance;

- one quality-related measure of your own.

You do not need a large history in the demo database. The purpose is to define how repeated Odoo Purchase/Receipt transactions could generate the metric over time.

Part F — Complete Tutorial 7 research.

Group 1: research a manufacturing company that successfully implemented SCM, whether ERP existed before SCM, the challenges, and how success was measured.

Group 2: compare CRM and SCM, including similarities, differences, interaction points, and technologies.

Use the SCM metrics from Chapter 7 to strengthen the Group 1 analysis and the demand-to-fulfilment hand-off to strengthen the Group 2 comparison.

Lab deliverable

Submit/share the Tutorial 7 presentation and reflection as instructed. Include Odoo evidence of one vendor-product link, one confirmed Purchase Order, and one Done Receipt. Your reflection should connect at least one research point back to the Odoo supply-chain process.

PRACTICAL MILESTONE

Lab 8 of 8

Human Resources and Recruitment Process Modelling

Use Odoo organisation data, then model the ABC recruitment process

Coverage. Tutorial 8 + Chapters 9–10.

Core system. Odoo Employees + process modelling.

Goal. Connect structured employee/department data to an end-to-end recruitment workflow and identify where information systems reduce delay and inconsistency.

How the practical connects to the concept

The supplied Odoo build guide uses Employees rather than a full recruitment workflow. Therefore the Odoo component focuses on organisation structure and operational ownership, while the Tutorial 8 recruitment scenario is modelled as a process. If the class deployment provides additional recruitment functionality, it may be demonstrated as an extension, but it is not required by the supplied guide.

Part A — Verify the organisation structure.

Open the four departments created in Lab 1. Check managers, employee headcount, and one organisation chart. Open the Sales Manager, Purchasing Officer, Production Planner, and Accountant records and confirm the linked user where applicable.

Part B — Complete the “dream job” activity.

Choose a role such as ERP Consultant, System Analyst, Database Administrator, or Business Development Manager. List:

- information needed to judge suitability;
- required technical skills;
- required behavioural/communication skills;
- evidence used to verify each requirement.

Part C — Diagnose the ABC recruitment process.

Identify candidate-experience risks in the scenario. Consider delays in scheduling, repeated requests for information, inconsistent job criteria, lost feedback, weak status communication, and errors in offer details. For each risk, state what an effective information system would change.

Part D — Draw a basic flowchart.

Start with “Department identifies need” and end with “Candidate accepts offer.” Include at least one decision such as “Candidate passes screening?” or “Candidate accepts offer?”

Part E — Draw a swimlane diagram.

Use lanes for at least:

- Hiring Manager / Marketing Department;
- Human Resources;
- Candidate;
- Interview/assessment stakeholders.

Mark hand-offs and identify at least three places where a shared workflow could reduce waiting.

Common mistake

A swimlane is not just the same flowchart with decorative horizontal lines. Every activity must sit in the lane of the role responsible for performing it, and arrows crossing lanes should represent real hand-offs.

Part F — Connect people to ERP transactions.

Using prior labs, identify where each operational role appears:

- Sales Manager → opportunity / quotation / Sales Order;
- Purchasing Officer → RFQ / Purchase Order;
- Production Planner → Manufacturing Order;
- Accountant → invoice/payment process.

Explain why this is more useful than creating employee names that never appear in business transactions.

Lab deliverable

Submit the Tutorial 8 answers, the flowchart, and the swimlane diagram as instructed by the tutor. Include one Odoo screenshot showing the department/employee structure and one short paragraph explaining the difference between an HR employee record and an operational user/login.

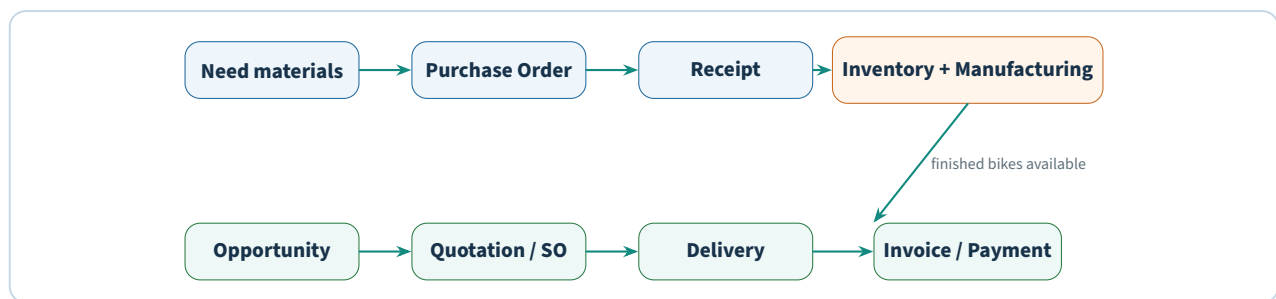
Capstone: West End Bicycles End-to-End ERP Story

The supplied Odoo guide is designed as one connected storyline. The strongest final demonstration is therefore not eight isolated screenshots, but one narrative showing how information and stock move across the company.

Source basis

This capstone follows the 21-step West End Bicycles Odoo 19 Community build guide supplied with Tutorial 1. It preserves the guide's company, roles, products, vendors, customer, Road Bike BOM, and RM 5,000 customer order.

The two connected business cycles



The buying cycle fills the warehouse; the selling cycle empties it. Manufacturing connects raw materials to finished goods. Finance closes the loop when money is received.

Capstone data set

Master data	Values used in the guide
Company	West End Bicycles, Ipoh, Perak, Malaysia
Finished product	Road Bike, Goods, Track Inventory, selling price RM 1,000
Road Bike BOM	1 Aluminium Frame, 2 Rubber Tires, 1 Brake Set, 1 Gear Set, 1 Bicycle Chain
Customer	Seri Iskandar Cycling Club
Customer order	5 Road Bikes, total expected/order value RM 5,000
Key operational roles	Sales Manager, Purchasing Officer, Production Planner, Accountant

The 21-step build roadmap

1. Create or access the Odoo database.
2. Install the eight required apps.
3. Set up West End Bicycles company profile.
4. Create departments, employees, managers, and selected operational users.

5. Create finished products.
6. Create raw-material products.
7. Create suppliers/vendors.
8. Link vendors to the raw materials they supply.
9. Create RFQs and confirm Purchase Orders.
10. Validate receipts so raw-material stock increases.
11. Create the Road Bike BOM.
12. Complete a Manufacturing Order for 5 Road Bikes.
13. Create Seri Iskandar Cycling Club as customer.
14. Create the CRM opportunity.
15. Move the opportunity through the pipeline.
16. Create and confirm the quotation/Sales Order for 5 Road Bikes.
17. Validate the Delivery so finished-bike stock decreases.
18. Create and post the customer invoice.
19. Register the customer payment.
20. Create/describe the cross-functional customer-order team.
21. Explain the information flow and show management reporting.

What should happen to stock

The physical logic is simple enough to use as a diagnostic tool.

Event	Raw-material stock	Road Bike stock
Validate vendor receipts	Increases	No change
Complete MO for 5 bikes	Decreases according to BOM	Increases by 5
Validate customer delivery	No change	Decreases by 5

If the stock movement does not follow this pattern, trace the document states upstream. A Purchase Order that was never received cannot supply Manufacturing; an MO that is not Done cannot create finished stock; a Sales Order cannot deliver bikes that do not exist on hand.

Who owns what

Role	Odoo ownership field	Business meaning
Sales Manager	Salesperson / Team Leader	Owns customer opportunity and commercial order
Purchasing Officer	Buyer	Owns vendor sourcing and purchase documents
Production Planner	Responsible	Owns the Manufacturing Order
Accountant	Finance/payment activity	Owns or supports invoicing and payment recording

The information-flow narrative

A polished demonstration can use this sequence:

1. **Customer demand enters CRM.** The opportunity records who wants what and the expected value.
2. **Sales formalises the deal.** The quotation records product, quantity, customer, price, and salesperson.
3. **Sales Order creates operational demand.** Inventory can now see the fulfilment requirement.
4. **Inventory confirms physical movement.** Delivery status proves whether goods actually left the warehouse.
5. **Finance reuses the sales data.** The invoice is generated from the order rather than typed from scratch.
6. **Payment closes the loop.** Sales and management can see settlement status.
7. **Reports aggregate the history.** Graph, list, and pivot views turn transaction records into management information.

Key idea

The capstone is successful when you can explain *why* each record exists, what event changes its state, what downstream record it enables, and which department needs the resulting information. A sequence of screenshots without that explanation is not yet an ERP demonstration.

Recommended final screenshots

The supplied guide recommends evidence across the entire process. A compact final set is:

1. installed apps and company profile;
2. departments/employees/users;
3. Road Bike product with Goods + Track Inventory;
4. raw material with linked vendor;
5. confirmed Purchase Order;
6. Done Receipt;
7. Road Bike BOM;
8. Done Manufacturing Order;

9. CRM opportunity;
10. confirmed Sales Order;
11. Done Delivery;
12. Posted invoice;
13. Paid/In Payment invoice;
14. one Graph or Pivot report.

Check your understanding

1. If Road Bike stock remains zero after the MO, which upstream records should you inspect?
2. If the invoice price is different from the Sales Order, what ERP integration principle has failed?
3. Which capstone steps demonstrate the link between CRM, SCM, and Finance?

Appendix A: Odoo 19 Community Quick Reference

This appendix condenses the terminology and troubleshooting rules in the supplied West End Bicycles guide.

Navigation and record behaviour

Term	Meaning
App grid	Main launcher for Sales, Purchase, Inventory, Manufacturing, CRM, Employees, and other installed apps.
Breadcrumb	Clickable trail showing the current app/record path; use it to return to earlier records.
Smart button	Button above a record that jumps to related records such as Receipt, Delivery, or Invoices.
New	Starts a new record. Existing records are already editable in the Odoo 19 workflow described by the guide.
Cloud/save indicator	Appears when changes are unsaved; navigating away also auto-saves in the guide's workflow.
Graph / Pivot	Reporting views that summarise transaction data for analysis.

Important Odoo 19 product rule

The supplied guide states that older “Storable Product” terminology should be translated to:

Product Type = Goods + **Track Inventory = enabled**

If Track Inventory is not enabled, Odoo will not maintain the stock count expected by the course workflow.

Core transaction vocabulary

Term	Course meaning
RFQ	Draft purchasing document before supplier commitment.
Purchase Order	Confirmed order to a vendor; creates/links a Receipt.
Receipt	Inventory transfer bringing purchased goods into stock.
BOM	Component recipe for one finished product.
Manufacturing Order	Instruction to produce a quantity; consumes components and produces finished stock.
Opportunity	Potential sale in CRM.
Quotation	Draft commercial offer to a customer.
Sales Order	Confirmed customer order; links to fulfilment and billing.
Delivery	Inventory transfer sending goods to the customer.
Invoice	Customer billing document; Draft before posting, then Posted, then Paid/In Payment after settlement.
On Hand	Quantity physically recorded in stock at the current moment.
Validate	Complete an inventory transfer and update stock.

Troubleshooting by process state

Problem	Likely cause	What to check
No Track Inventory option / stock not moving	Product setup or Inventory app issue	Confirm Inventory is installed, Product Type is Goods, and Track Inventory is enabled.
No New Quotation from CRM	Opportunity/customer/app state incomplete	Save opportunity, set customer, confirm Sales is installed; otherwise create quotation from Sales.
Vendor does not appear on product purchasing	Vendor record exists but is not linked to product	Product Purchase tab, vendor line, price, and Can be Purchased.
Cannot receive goods	RFQ not yet confirmed	Confirm RFQ into Purchase Order, then open Receipt.
Cannot manufacture	Components are not actually on hand	Validate supplier receipts and check component On Hand.
Cannot deliver	Finished product is unavailable	Complete Manufacturing Order and verify Road Bike On Hand.
Cannot register payment	Invoice still Draft	Post/confirm the invoice before Register Payment.
Expected button missing	Record is in the wrong lifecycle state	Check whether the prior business event is Draft, Confirmed/Posted, or Done.

Key idea

When troubleshooting, follow the process backwards. A downstream action is usually unavailable because an upstream business event has not been confirmed or completed.

Appendix B: Course Source and Tutorial Map

These notes were built from the supplied tutorial documents, the Odoo West End Bicycles build guide, the supplied forecast/S&OP spreadsheet, and the provided third edition of *Concepts in Enterprise Resource Planning*. The table below shows how the materials were aligned.

Tutorial	Main activity	Notes / textbook support
T1	Business functions, information flows, process design	Chapter 1 of the textbook; Odoo guide provides West End Bicycles organisation and the full integrated process.
T2	ERP benefit and implementation challenge	Chapter 2 plus implementation/change-management concepts from Chapter 7.
T3	Production approach, forecast, S&OP	Chapter 4 production-planning hierarchy; Odoo BOM and Manufacturing Order steps.
T4	CRM proposal for Luna Electronics	Chapter 3 CRM section; Odoo CRM pipeline and quotation conversion.
T5	Sales Order Process / Fitter Snacker questions	Chapter 3 Fitter Snacker sales problems, ERP sales order processing, and Sales/Accounting integration.
T6	Footloose forecast and S&OP spreadsheet	Chapter 4 forecasting and S&OP formulas; supplied spreadsheet values.
T7	SCM research and CRM-vs-SCM comparison	Chapter 4 traditional/integrated supply chain and SCM measures of success; Odoo Purchase/Inventory flow.
T8	HR recruitment and process modelling	Chapter 6 recruiting/hiring problems + Chapter 7 flowcharts/swimlanes; Odoo Employees organisation records.

Source discrepancies preserved in the notes

The materials contain a few inconsistencies that students should recognise rather than have silently corrected for them:

- The Tutorial 6 Word file asks students to forecast 2023 from 2022 sales, while the supplied spreadsheet headings still refer to 2022.
- Tutorial 6 states safety stock = 100; the spreadsheet pre-fills 500 in the initial December inventory cell.
- Tutorial 6 describes sales in pairs of shoes but production capacity as 40 shoes per hour, creating a unit-of-measure ambiguity.
- Tutorial numbering in some source documents is inconsistent (for example, a page header may use a different tutorial number from the title). The notes organise topics by subject rather than reproducing those numbering mismatches.

Key idea

When a source file and a spreadsheet disagree, do not hide the conflict. State the assumption used, keep units explicit, and confirm assessment expectations with the tutor.